Circuit Theorems Assignment

ECE2107 Circuits & Systems · assignment marathon · built so far: Part A Superposition (5.1–5.14), Thevenin & Norton (5.15–5.34)

How to use this

Every question follows the same order:

1. **Question**, with the figure taken straight from the book.
2. **Try it first** (yellow). Cover the rest of the question and solve it.
3. **Hint: what's new** (orange). Only when the question brings an idea you haven't met earlier in this PDF. If you're stuck, read it and try again.
4. **Solution** in compact assignment form: Given → To find → each step with its intermediate values → boxed answer. Copy it into your submission, or compare it line by line with your attempt to find the step that went wrong.
5. **Side note** (grey) or **book error** (red), only when there is something worth knowing.

The tag next to each question number shows how much teaching it carries: **teach** full explanation · **new idea** one new thing · **practice** plain solution.

If you couldn't solve one, just tell me its number. Nothing else needs marking.

What's in it (full version)

Part	Book	Questions	Count
A	Ghosh & A K Chakraborty, *Circuit Theory and Networks*, ch. 5 (**solved**)	Superposition 5.1–5.14 · Thevenin/Norton 5.15–5.34 · Max power 5.35–5.44	44
B	Alexander & Sadiku, *Fundamentals of Electric Circuits* 5e (**unsolved**)	Practice problems + end-of-chapter: ch. 4 (superposition, source transformation, Thevenin/Norton, max power), ch. 10 (the same in AC), ch. 11.3 (AC max power)	107 + practice problems

Scope follows the official MUJ ECE2107 syllabus: superposition, Thevenin's and Norton's theorems, maximum power transfer, DC and AC, plus source transformation.

Checking. Every answer was worked out again and confirmed by a full circuit solve with all sources on (`src/verify.py`), including each case's partial answer. Matches with the book are marked "= book". Where the book is wrong, the correct answer is given and a red box explains the book's slip.

Part A · Chakraborty (solved problems)

Ghosh & A K Chakraborty, *Circuit Theory and Networks* 4e, ch. 5 "Network Theorems", Solved Problems

A · Superposition theorem

KVL, KCL, Ohm's law (you have these) → Nodal / mesh: all sources at once → **Superposition: one source at a time** → Thevenin / Norton (next)

Where this sits. You can already solve any circuit with nodal or mesh analysis, taking every source at once. Superposition breaks the same job into smaller circuits, **one source at a time**, which is often quicker by hand. It is also the reason Thevenin's theorem works, which comes next.

Three terms first.

- **Linear circuit:** made of R, L, C, linear dependent sources and independent sources. Doubling every source doubles every current and voltage.
- **Independent source:** a fixed voltage or current (a battery, a 2 A source, 10 cos 4t).
- **Dependent source** (diamond symbol): its value follows some other voltage or current in the circuit, for example $2V_x$.

Statement. In a linear circuit with several independent sources, any current or voltage is the **algebraic sum** of the values caused by **each independent source acting alone**.

Switching a source off means setting its value to zero:

Source	Switched off it becomes	Why
Ideal voltage source	**short circuit**	0 V across it
Ideal current source	**open circuit**	0 A through it
Dependent source	**never switched off**	it is not a source on its own; it follows the circuit

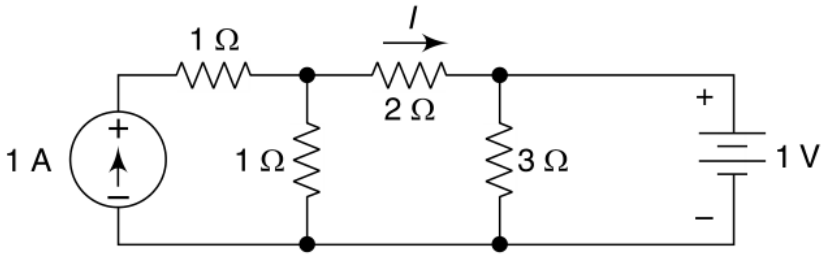
Superposition adds currents and voltages only, **not power** (P goes as I^2, and $(I' + I'')^2 \neq I'^2 + I''^2$).

A1 · Problem 5.1

Superposition · DC **teach**

QUESTION

Find the current I in the circuit shown, using the superposition theorem.



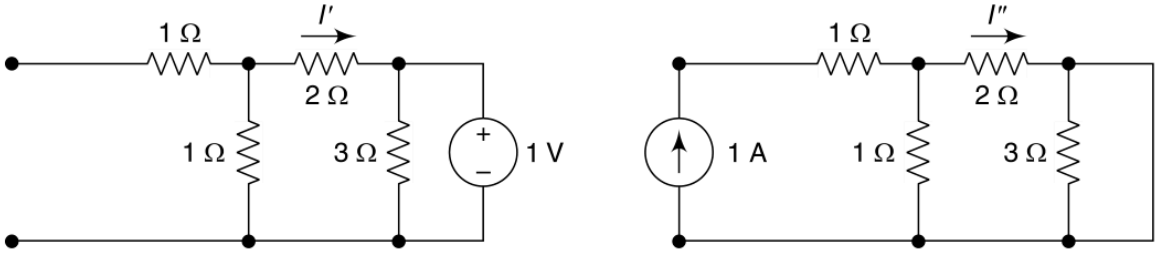
I flows left to right through the 2 Ω resistor.

TRY IT FIRST

Cover everything below. Draw two circuits, 1 V alone and 1 A alone, find the current in the 2 Ω in each, and decide each one's sign before adding.

HINT: WHAT'S NEW

- **The whole method:** two sources, so solve **two small circuits** and add.
- **Switching off:** the 1 V source becomes a short; the 1 A source becomes an open.
- **Signs:** measure each part against the arrow drawn for I. A part flowing the other way is negative.



(i) 1 V acting alone (1 A opened) · (ii) 1 A acting alone (1 V shorted)

Given: 1 A source, 1 V source; 1 Ω (series), 1 Ω (shunt), 2 Ω, 3 Ω.

To find: I through the 2 Ω (left → right).

Case (i): 1 V alone (1 A opened). The series 1 Ω carries nothing. The 3 Ω sits directly across the 1 V, so it changes nothing else. The 1 V drives current through 2 Ω + 1 Ω (shunt):

$$\text{current} = \frac{1}{2+1} = \frac{1}{3}\ \text{A (right → left)} \qquad \Rightarrow \qquad I' = -\frac{1}{3}\ \text{A}$$

Case (ii): 1 A alone (1 V shorted, so the 3 Ω is shorted out). The 1 A divides between the 1 Ω shunt and the 2 Ω:

$$I'' = 1 \times \frac{1}{1+2} = \frac{1}{3}\ \text{A (left → right)}$$

Superposition: $I = I' + I'' = -\frac{1}{3} + \frac{1}{3} = 0$

ANSWER

I = 0 A (= book)

WHY ZERO, AND TWO SHORTCUTS

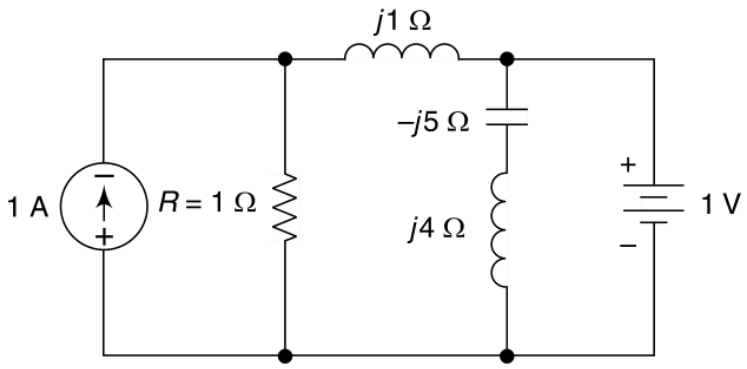
- Node check: $1 = \frac{V_A}{1} + \frac{V_A - 1}{2}$ gives $V_A = 1$ V, the same as the right-hand node, so no current flows in the 2 Ω.
- An element **in series with an ideal current source** never changes any other current (the left 1 Ω here).
- An element **in parallel with an ideal voltage source** never changes any other current (the 3 Ω here).

A2 · Problem 5.2

Superposition · AC **teach**

QUESTION

Calculate the voltage V across the resistor R by using the superposition theorem.



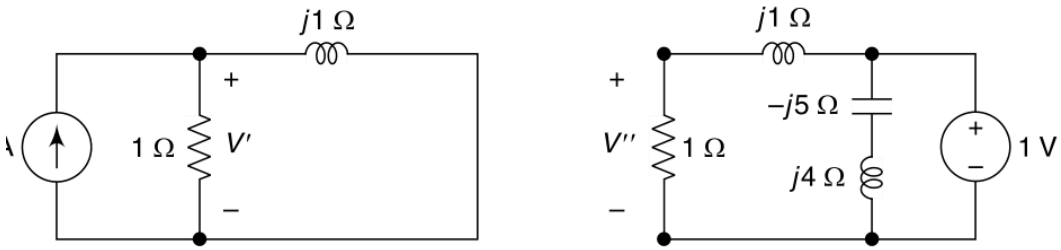
The book draws +/− signs inside the current source. Read it by its arrow: 1 A pushed up into the top node.

TRY IT FIRST

Which elements drop out in each case? Aim for V′ and V″ in the form $\frac{\text{something}}{1 + j}$.

HINT: WHAT'S NEW

- **AC (phasor) circuit.** j1 Ω is an inductor, −j5 Ω a capacitor. Superposition is unchanged; the arithmetic uses complex numbers.
- **Complex dividers:** the same formulas as for resistors, with Z in place of R.
- The −j5 + j4 branch is **across the 1 V source** (5.1's shortcut), so it never matters.



(i) 1 A acting alone · (ii) 1 V acting alone

Given: 1 A, 1 V; R = 1 Ω, j1 Ω, −j5 Ω, j4 Ω.

To find: V across R (+ at top).

Case (i): 1 A alone (1 V shorted; this also shorts out the −j5 + j4 branch). R ∥ j1 across the 1 A:

$$V' = 1 \times \frac{1 \times j1}{1 + j1} = \frac{j}{1 + j}\ \text{V}$$

Case (ii): 1 V alone (1 A opened). j1 and R in series across 1 V:

$$I'' = \frac{1}{1 + j} \qquad \Rightarrow \qquad V'' = 1 \times I'' = \frac{1}{1 + j}\ \text{V}$$

Superposition: $V = \frac{j}{1 + j} + \frac{1}{1 + j} = 1\ \text{V}$

ANSWER

V = 1∠0° V (= book)

CHECK

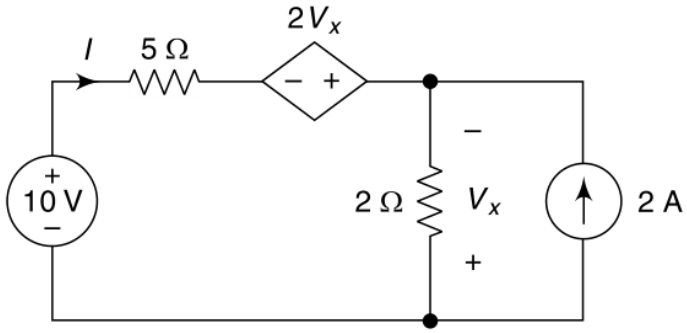
Node equation at the top node: $1 = \frac{V}{1} + \frac{V - 1}{j1}$ gives V(1 + j) = 1 + j, so V = 1 V ✓.

A3 · Problem 5.3

Superposition · dependent source **teach**

QUESTION

Use the superposition theorem on the circuit shown to find I.



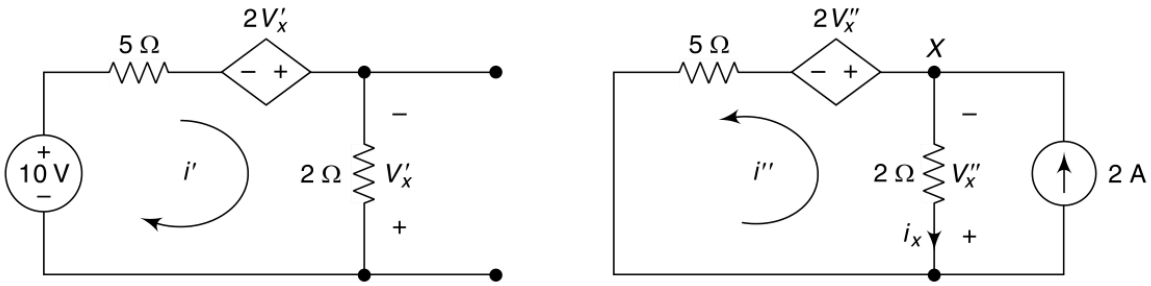
$2V_x$ is a voltage-controlled voltage source. V_x is across the 2 Ω with + at the bottom.

TRY IT FIRST

Write V_x' in terms of the loop current in case (i). For case (ii), write one KCL at the top of the 2 Ω and one KVL around the left loop. Expect elevenths.

HINT: WHAT'S NEW

- **Dependent source:** it **stays in every case**. Only the 10 V and the 2 A take turns.
- **Its value changes per case:** $2V_x'$ in case (i), $2V_x''$ in case (ii). Re-find V_x in each sub-circuit.
- **Polarity of V_x:** + is at the **bottom** of the 2 Ω. Most wrong answers to this question come from this sign.



(i) 10 V alone (2 A opened) · (ii) 2 A alone (10 V shorted); the dependent source stays in both

Given: 10 V, 2 A, 5 Ω, 2 Ω, VCVS $2V_x$.

To find: I (out of the 10 V source, through the 5 Ω).

Case (i): 10 V alone. One loop, current i′ clockwise (= direction of I), flowing down the 2 Ω, so $V_x' = -2i'$.

$$10 + 2V_x' = 5i' + 2i' \quad \Rightarrow \quad 10 - 4i' = 7i' \quad \Rightarrow \quad I' = i' = \frac{10}{11}\ \text{A}$$

Case (ii): 2 A alone. At node X: 2 A splits into i_x (down the 2 Ω, so $V_x'' = -2i_x$) and i″ (leftward through the dependent source and 5 Ω).

$$\text{KCL at X:} \qquad 2 = -\frac{V_x''}{2} + i'' \qquad \ldots(1)$$

$$\text{Left loop:} \qquad 5i'' + 3V_x'' = 0 \qquad \Rightarrow \qquad i'' = -\frac{3}{5}V_x'' \qquad \ldots(2)$$

(X sits at $-V_x''$; the dependent source drops another $2V_x''$, so the 5 Ω has $-3V_x''$ across it.)

$$\text{(2) in (1):} \qquad 2 = -\frac{11}{10}V_x'' \qquad \Rightarrow \qquad V_x'' = -\frac{20}{11}\text{ V}, \qquad i'' = \frac{12}{11}\text{ A}$$

i'' runs anticlockwise, against I, so **$I'' = -12/11$ A**.

Superposition: $I = \frac{10}{11} - \frac{12}{11} = -\frac{2}{11}$ A

ANSWER

$I = -2/11$ A ≈ −0.182 A (= book). The minus sign means the current actually flows into the + of the 10 V source.

NEVER SWITCH OFF A DEPENDENT SOURCE

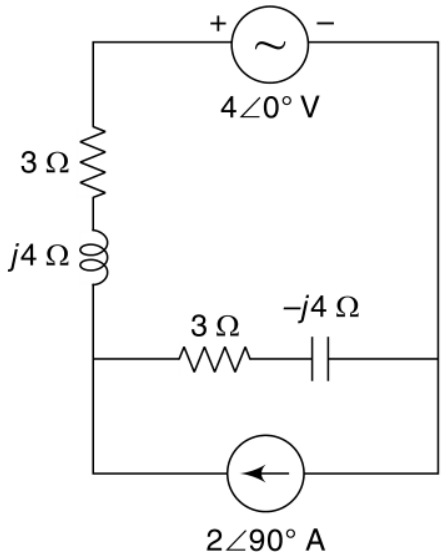
Shorting $2V_x$ as if it were independent gives a wrong answer. It stays in; only its value changes from case to case.

A4 · Problem 5.4

Superposition · AC **new idea**

QUESTION

Determine the current in the capacitor branch by the superposition theorem.



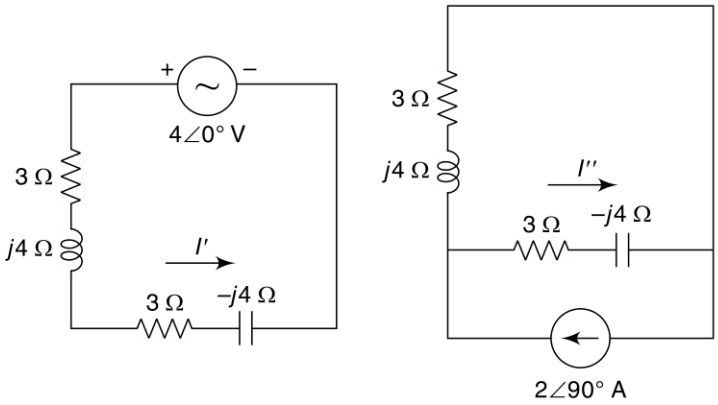
$Z_1 = 3 + j4\ \Omega$ (left branch); $Z_2 = 3 - j4\ \Omega$ (capacitor branch); current source $2\angle 90°$ A.

TRY IT FIRST

Which way does each source see Z_1 and Z_2: in series, or in parallel?

HINT: WHAT'S NEW

- **Source with an angle:** $2\angle 90°$ A = j2 A. Convert to rectangular form before multiplying.
- **Polar form with a negative real part:** a calculator's $\tan^{-1}$ lands in the wrong quadrant; add 180°.



(i) $4\angle 0°$ V alone · (ii) $2\angle 90°$ A alone

Given: $4\angle 0°$ V, $2\angle 90°$ A, $Z_1 = 3 + j4\ \Omega$, $Z_2 = 3 - j4\ \Omega$.

To find: I in the capacitor branch (left → right).

Case (i): $4\angle 0°$ V alone (current source opened). Z_1, Z_2 in series:

$$I' = \frac{4\angle 0°}{(3 + j4) + (3 - j4)} = \frac{2}{3}\angle 0°\ \text{A}$$

Case (ii): 2∠90° A alone (voltage source shorted). $Z_1 \parallel Z_2$ across the source:

$$I'' = j2 \times \frac{3 + j4}{6} = -\frac{4}{3} + j1 \text{ A}$$

Superposition: $I = \frac{2}{3} - \frac{4}{3} + j1 = -\frac{2}{3} + j1$ A

$$|I| = \sqrt{((2/3)^2 + 1)} = 1.20 \text{ A}, \qquad \theta = 180° - \tan^{-1}(1.5) = 123.7°$$

ANSWER

I = 1.2∠123.7° A (= book)

SIDE NOTE

$Z_1 + Z_2 = 6\ \Omega$ is purely real (+j4 and −j4 cancel), which is why both cases are clean. Equal and opposite reactances in series act like a short: **series resonance**, met later.

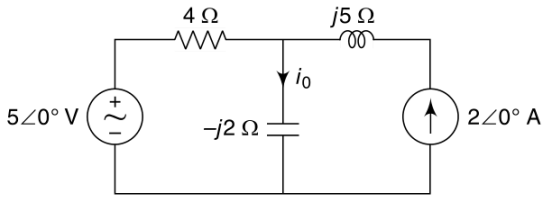
A5 · Problem 5.5

Superposition · AC, DC + AC **teach**

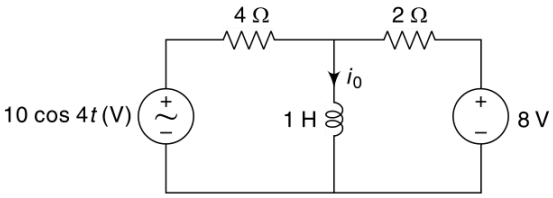
QUESTION

Find the current i_0 using the superposition theorem, in each of circuits (a), (b) and (c).

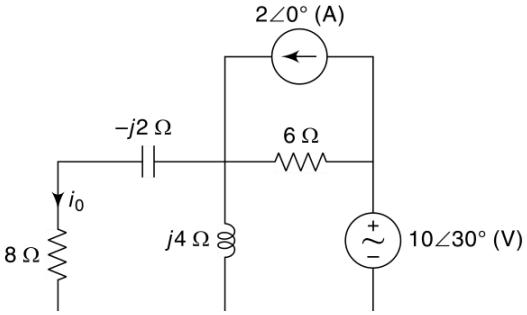
(a)



(b)



(c)



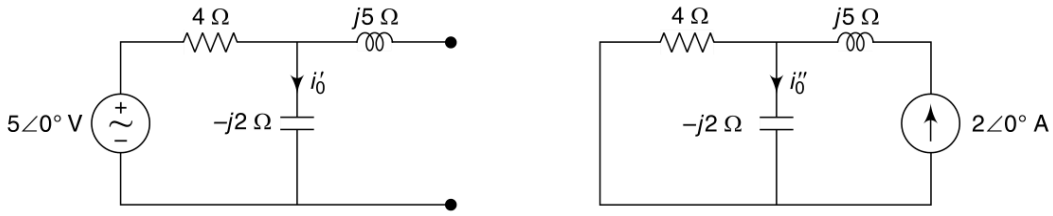
TRY IT FIRST

(a) is quick practice. In (b), stop and ask: what does an inductor look like to the 8 V **DC** source? In (c), redraw the circuit after switching off each source before you calculate anything.

HINT: WHAT'S NEW

- **(b) Two different frequencies (DC and 10 cos 4t).** Each case is solved **at its own frequency**: at DC ($\omega = 0$), $X_L = \omega L = 0$, so the **inductor is a short**; at $\omega = 4$, it is j4 Ω. The two answers are added **in the time domain**, never as phasors, because phasors only add when they share one frequency.
- **(c) Redraw after switching off.** Once the voltage source is shorted, the 6 Ω and the current source both run to ground and everything becomes parallel.

(a) Given: 5∠0° V, 2∠0° A, 4 Ω, −j2 Ω, j5 Ω. **To find:** i_0 (down the capacitor).



(i) voltage source alone · (ii) current source alone

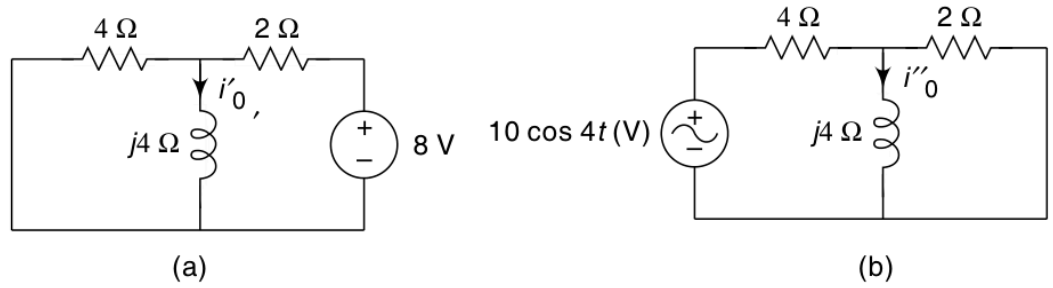
$$5\angle 0° \text{ alone:} \qquad i_0' = \frac{5}{4 - j2} = 1 + j0.5 \text{ A}$$

$$2\angle 0° \text{ alone (j5 is in series with the source, so irrelevant):} \qquad i_0'' = 2 \times \frac{4}{4 - j2} = 1.6 + j0.8 \text{ A}$$

ANSWER

(a) $i_0 = 2.6 + j1.3 = 2.91\angle 26.57°$ A (= book, 2.9∠26.56° A)

(b) Given: 10 cos 4t V, 8 V DC, 4 Ω, 1 H, 2 Ω. **To find:** $i_0(t)$ (down the inductor).



The book's sub-circuits (a) DC alone, (b) AC alone. In (a) the inductor must be read as a short circuit.

8 V DC alone (AC source shorted; $\omega = 0$, so the inductor is a short):

$$i_0' = \frac{8}{2} = 4\ \text{A} \qquad \text{(the 4 Ω has 0 V across it)}$$

10 cos 4t alone (8 V shorted; ω = 4 rad/s, Z_L = j4 Ω):

$$Z = 4 + \frac{j4 \times 2}{2 + j4} = 5.6 + j0.8\ \Omega, \qquad I = \frac{10\angle 0°}{Z} = 1.75 - j0.25\ \text{A}$$

$$i_0'' = I \times \frac{2}{2 + j4} = 0.25 - j0.75 = 0.791\angle -71.57°\ \text{A}$$

Superposition (time domain):

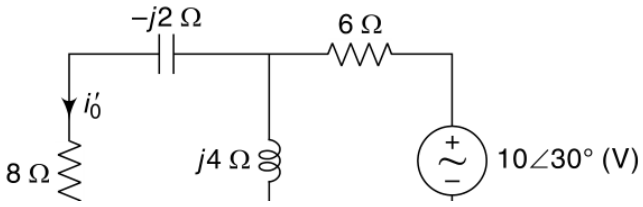
ANSWER

(b) $i_0(t) = 4 + 0.791 \cos(4t - 71.57°)$ A

BOOK ERROR IN 5.5(B)

The book treats the inductor as j4 Ω for the **DC** source too, and then adds the DC and AC results as if both were phasors, printing 2.63∠−63.58° A. Both steps are wrong: at DC an inductor is a short, and quantities at different frequencies cannot be added as phasors. The answer above was confirmed by a node solve of each case.

(c) Given: 10∠30° V, 2∠0° A, 8 Ω, −j2 Ω, j4 Ω, 6 Ω. **To find:** i_0 (down the 8 Ω).

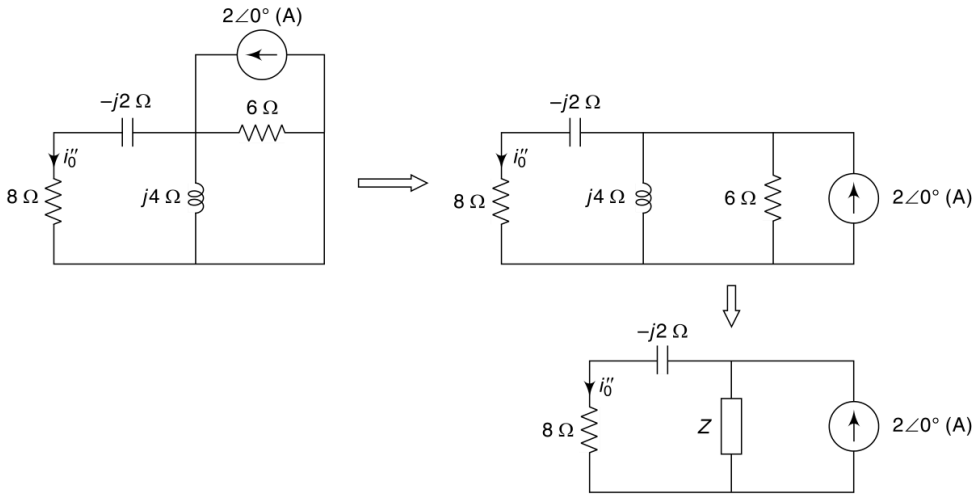


(i) voltage source alone

10∠30° V alone (current source opened):

$$Z = 6 + j4 \parallel (8 - j2) = 6 + (1.882 + j3.529) = 7.882 + j3.529\ \Omega, \qquad I = \frac{10\angle 30°}{Z} = 1.158\angle 5.88°\ \text{A}$$

$$i_0' = I \times \frac{j4}{(8 - j2) + j4} = I \times \frac{j4}{8 + j2} = 0.562\angle 81.84°\ \text{A}$$



(ii) current source alone, redrawn: the 6 Ω and the source now both go to ground

2∠0° A alone (voltage source shorted). Z = j4 ∥ 6 = 1.846 + j2.769 Ω:

$$i_0'' = 2 \times \frac{Z}{(8 - j2) + Z} = \frac{j24}{28 + j22} = 0.674\angle 51.84^\circ \text{ A}$$

$$i_0 = (0.080 + j0.556) + (0.416 + j0.530) = 0.496 + j1.086 \text{ A}$$

ANSWER

(c) $i_0 = 1.19\angle 65.45^\circ$ A

BOOK ERROR IN 5.5(C)

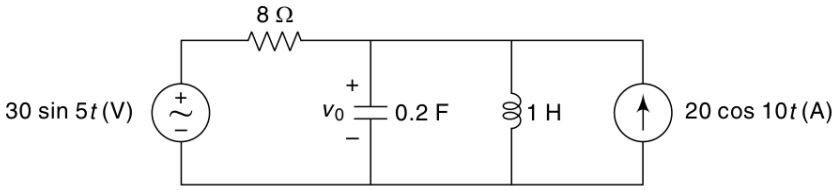
The book prints 0.81∠39.38° A. Two slips: in case (i) it divides the current with (8 − j2)/(8 + j2) instead of j4/(8 + j2) (the current into a branch is set by the **other** branch's impedance); in case (ii) it expands (8 − j2)(3 + j2) + j12 as 12 + j11 instead of 28 + j22. The answer above was confirmed by a full node solve.

A6 · Problem 5.6

Superposition · two frequencies **new idea**

QUESTION

Find v_0 using the superposition theorem.

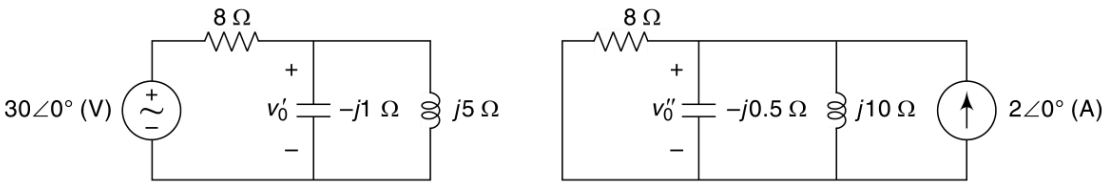


TRY IT FIRST

Work out X_C and X_L separately for ω = 5 and ω = 10 before drawing anything.

HINT: WHAT'S NEW

- **Reactances change with frequency.** $X_C = \frac{1}{\omega C}$ and $X_L = \omega L$ must be recomputed for each source's own ω.
- **Mixed sin and cos:** keep each term in the function it started with (sin for the voltage source, cos for the current source).



(i) voltage source alone, ω = 5 · (ii) current source alone, ω = 10

Given: 30 sin 5t V with 8 Ω; 0.2 F, 1 H, 20 cos 10t A. **To find:** $v_0(t)$.

30 sin 5t alone (ω = 5; current source opened): $X_C = \frac{1}{5 \times 0.2} = 1\ \Omega \to -j1\ \Omega$; $X_L = 5 \times 1 = 5\ \Omega \to j5\ \Omega$.

$$\frac{30 - v'}{8} = \frac{v'}{-j1} + \frac{v'}{j5} \quad \Rightarrow \quad v' = \frac{30}{8(0.125 + j0.8)} = 4.631\angle -81.12°\ \text{V}$$

20 cos 10t alone (ω = 10; voltage source shorted): $X_C = \frac{1}{10 \times 0.2} = 0.5\ \Omega \to -j0.5\ \Omega$; $X_L = 10\ \Omega \to j10\ \Omega$.

$$20 = v''\left(\frac{1}{8} + \frac{1}{-j0.5} + \frac{1}{j10}\right) = v''(0.125 + j1.9) \quad \Rightarrow \quad v'' = 10.50\angle -86.24°\ \text{V}$$

ANSWER

$v_0(t) = 4.631 \sin(5t - 81.12°) + 10.50 \cos(10t - 86.24°)$ V

BOOK INCONSISTENCY IN 5.6

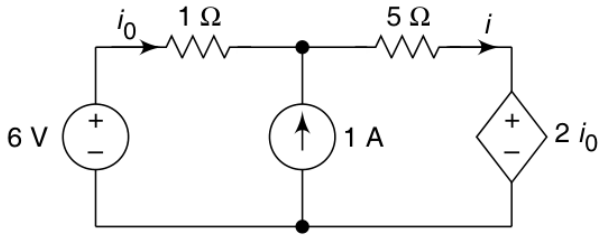
The figure gives the current source as **20** cos 10t A, but the book's solution uses **2**∠0° A and prints 1.051 cos(10t − 86.24°) for the second term. The answer above follows the figure. If your source is 2 A, the second term is 1.051 cos(10t − 86.24°) V; the first term is unchanged.

A7 · Problem 5.7

Superposition · current-controlled source **new idea**

QUESTION

Find i_0 and i from the circuit of the figure using the superposition theorem.



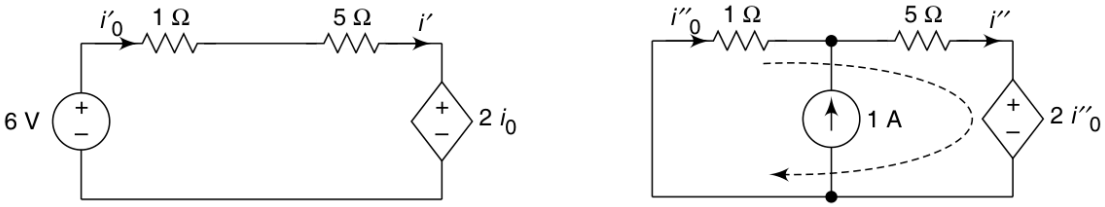
$2i_0$ is a current-controlled voltage source.

TRY IT FIRST

In each case, the dependent source's value is 2 × (that case's i_0). Write KCL at the middle node and one KVL around the outer loop.

HINT: WHAT'S NEW

- **Controlled by a current, not a voltage** (5.3 was voltage-controlled). The rule is the same: keep the source, and re-find its controlling current i_0 in each case.



(i) 6 V alone · (ii) 1 A alone

Given: 6 V, 1 A, 1 Ω, 5 Ω, CCVS $2i_0$. **To find:** i_0, i.

6 V alone (1 A opened): one loop, so $i_0' = i'$.

$$6 = 1\cdot i' + 5\cdot i' + 2i_0' = 8i' \quad \Rightarrow \quad i' = i_0' = 0.75 \text{ A}$$

1 A alone (6 V shorted):

$$\text{KCL:} \quad i'' = 1 + i_0'' \qquad \text{outer KVL:} \quad 1\cdot i_0'' + 5i'' + 2i_0'' = 0$$

$$\Rightarrow \quad 3i_0'' + 5(1 + i_0'') = 0$$

$$8i_0'' = -5 \quad \Rightarrow \quad i_0'' = -0.625 \text{ A}, \quad i'' = 0.375 \text{ A}$$

ANSWER

$i = 0.75 + 0.375 = 1.125$ A, $i_0 = 0.75 - 0.625 = 0.125$ A

BOOK ERROR IN 5.7

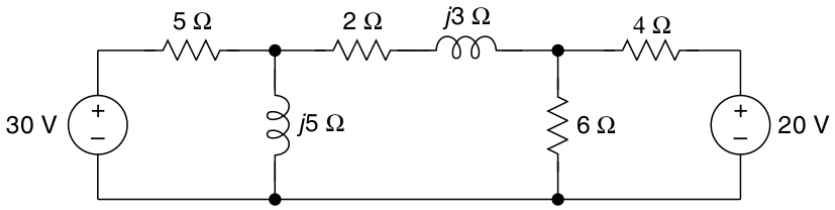
The book sets up the same equation, then solves $8i_0'' = -5$ as $i_0'' = -5/4 = -1.25$ A, and prints $i = 0.5$ A, $i_0 = -0.5$ A. The correct value is $-5/8 = -0.625$ A. Direct check: KCL $i = i_0 + 1$ and KVL $6 - i_0 - 5i = 2i_0$ give $i_0 = 1/8$ A ✓.

A8 · Problem 5.8

Superposition · AC **practice**

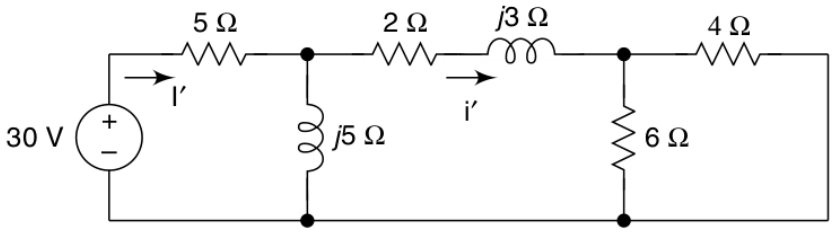
QUESTION

Using the superposition theorem, calculate the current through the (2 + j3) Ω impedance branch.



TRY IT FIRST

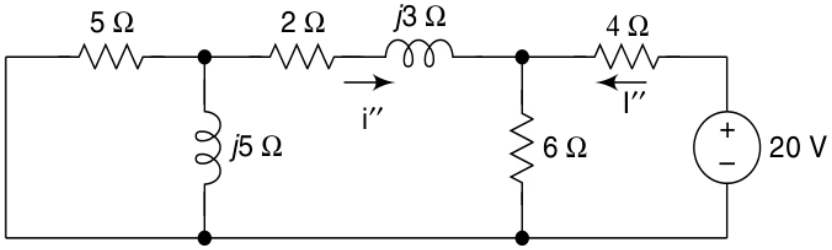
Series-parallel reduction, twice. Watch which way the branch current flows in case II.



Case I: 30 V alone

Case I: 30 V alone (20 V shorted). Branch: $Z_b = (2 + j3) + (6 \parallel 4) = 4.4 + j3$ Ω.

$$Z = 5 + \frac{Z_b \times j5}{Z_b + j5} = 6.32 + j2.60\ \Omega, \qquad I' = \frac{30}{Z} = 4.06 - j1.67\ \text{A}, \qquad i' = I' \times \frac{j5}{Z_b + j5} = 2.389 + j0.270\ \text{A}$$



Case II: 20 V alone

Case II: 20 V alone (30 V shorted). Branch: $Z_a = (5 \parallel j5) + 2 + j3 = 4.5 + j5.5$ Ω.

$$Z = 4 + \frac{6Z_a}{Z_a + 6} = 7.31 + j1.41\ \Omega, \qquad I'' = \frac{20}{Z} = 2.638 - j0.509\ \text{A}$$

$$i'' = -I'' \times \frac{6}{Z_a + 6} = -1.063 + j0.848\ \text{A} \qquad \text{(it flows right} \rightarrow \text{left, against i)}$$

ANSWER

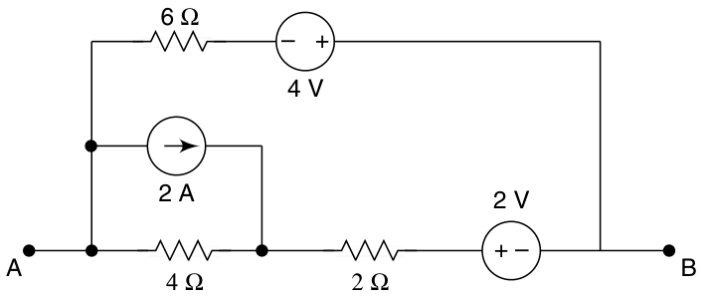
i = i′ + i″ = 1.326 + j1.117 = 1.73∠40.13° A (= book)

A9 · Problem 5.9

Superposition · three sources, open terminals **new idea**

QUESTION

Using the superposition theorem, find V_{AB}.

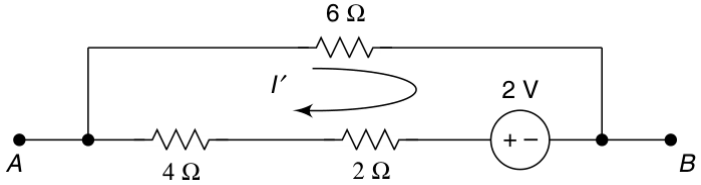


TRY IT FIRST

Three sources, so three cases. Nothing is connected across A–B, so no current leaves at A or B.

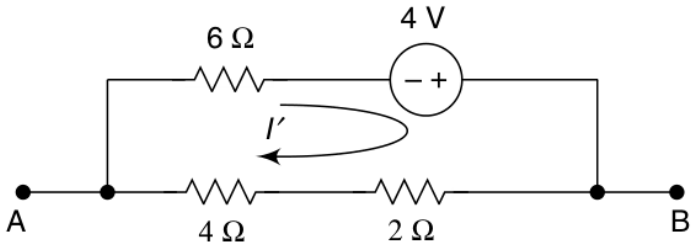
HINT: WHAT'S NEW

- **Three sources:** same method, one more case.
- **Open terminals:** V_{AB} is the voltage across a gap. With no load, the only path is the loop 4 Ω + 2 Ω + 6 Ω, and V_{AB} is the drop across the 6 Ω.
- **Source transformation inside a case:** 2 A in parallel with 4 Ω equals an 8 V source in series with 4 Ω, which makes case III a single loop.



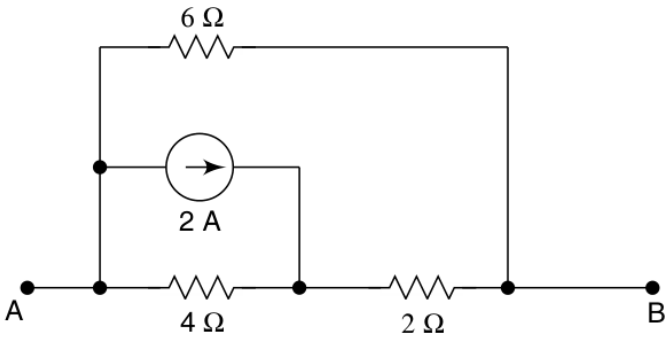
Case I: 2 V alone

Case I (2 V): $I' = \dfrac{2}{4+2+6} = \dfrac{1}{6}$ A, $V_{AB}' = 6I' = 1$ V

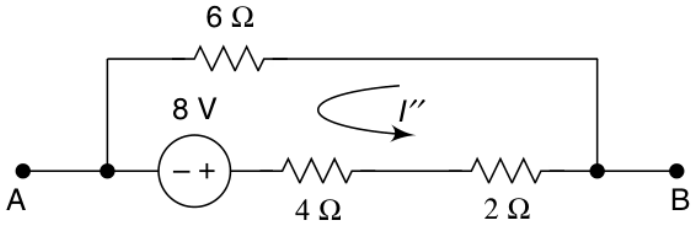


Case II: 4 V alone

Case II (4 V): $I'' = \dfrac{4}{12} = \dfrac{1}{3}$ A, $V_{AB}'' = -6I'' = -2$ V



Case III: 2 A alone



Case III after source transformation: 2 A ∥ 4 Ω → 8 V in series with 4 Ω

Case III (2 A): $I''' = \dfrac{8}{12} = \dfrac{2}{3}$ A, $V_{AB}''' = -6I''' = -4$ V

(In case I the current runs through the 6 Ω from A's side to B's side, so A is higher; in cases II and III it runs the other way, hence the minus signs.)

ANSWER

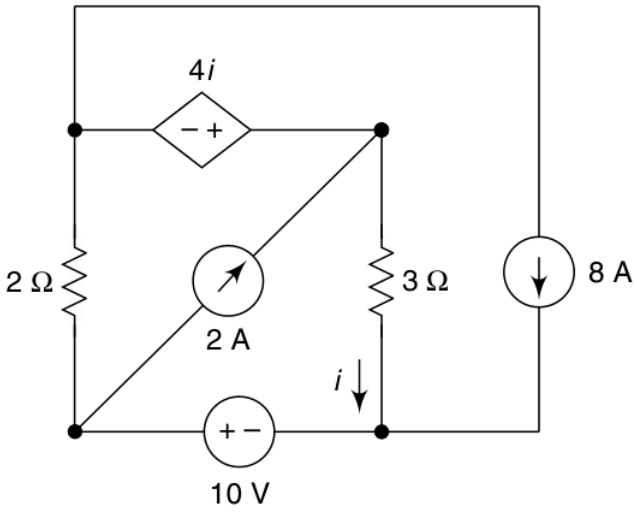
$V_{AB} = 1 - 2 - 4 = -5$ V (= book)

A10 · Problem 5.10

Superposition · dependent source, three sources **new idea**

QUESTION

Find the current i in the circuit shown using the superposition theorem.



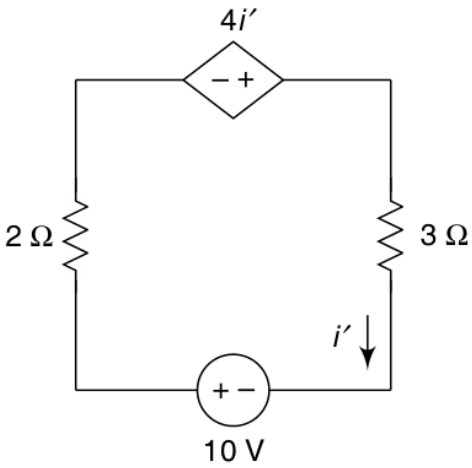
4i is a current-controlled voltage source; i flows down the 3 Ω.

TRY IT FIRST

Case I is one loop. In case II, look for a resistor in series with the dependent source.

HINT: WHAT'S NEW

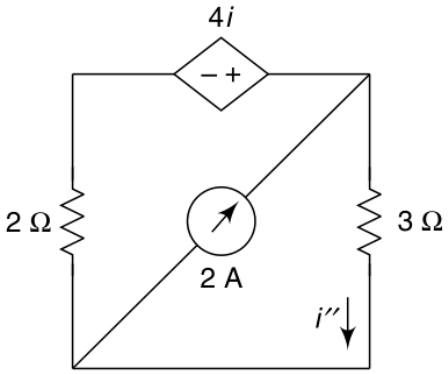
- **Source transformation works on dependent sources too.** In case II, the 4i source in series with the 2 Ω becomes a dependent current source 4i/2 = **2i** in parallel with 2 Ω.
- **A current source inside a mesh fixes that mesh current** (case III: the 8 A source sets the mesh difference I − 8).



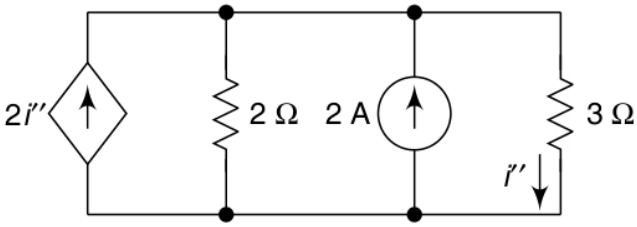
Case I: 10 V alone

Case I: 10 V alone (2 A and 8 A opened). Loop current i′ clockwise:

$$-4i' + 3i' - 10 + 2i' = 0 \qquad \Rightarrow \qquad i' = 10\ \text{A}$$



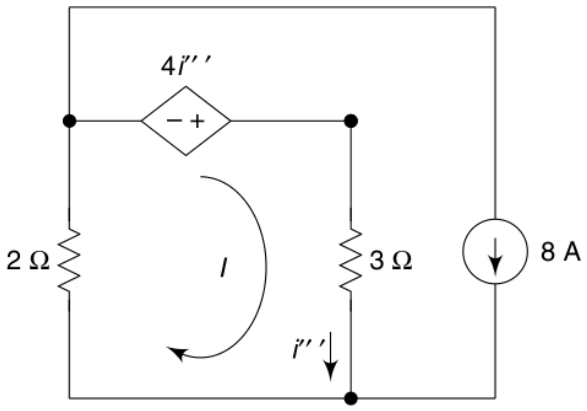
Case II: 2 A alone



Case II after converting 4i″ + 2 Ω into 2i″ ‖ 2 Ω

Case II: 2 A alone (10 V shorted, 8 A opened). Total (2 + 2i″) splits between 2 Ω and 3 Ω:

$$i'' = \frac{2}{2+3}(2 + 2i'') \qquad \Rightarrow \qquad 5i'' = 4 + 4i'' \qquad \Rightarrow \qquad i'' = 4\ \text{A}$$



Case III: 8 A alone

Case III: 8 A alone (10 V shorted, 2 A opened). Mesh current I; the 3 Ω carries i‴ = I − 8:

$$-4i''' + 3(I - 8) + 2I = 0, \qquad I = i''' + 8 \qquad \Rightarrow \qquad -4i''' + 3i''' + 2i''' + 16 = 0$$
$$\Rightarrow \qquad i''' = -16\ \text{A}$$

ANSWER

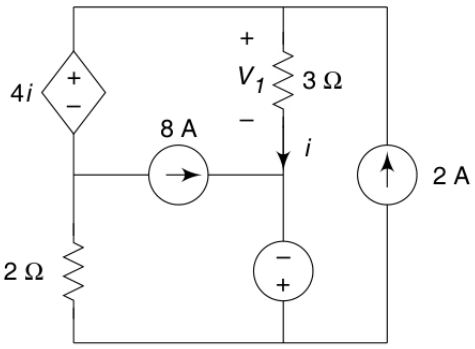
i = 10 + 4 − 16 = −2 A (= book)

A11 · Problem 5.11

Superposition · dependent source, three sources **practice**

QUESTION

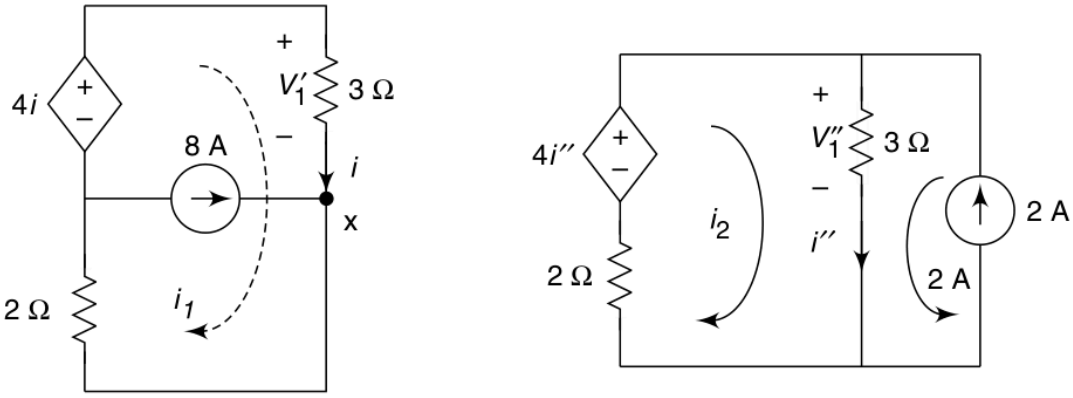
Using the superposition theorem, determine V_1, the voltage across the 3 Ω resistor.



The figure prints no value on the bottom-middle voltage source; the book's solution uses 10 V.

TRY IT FIRST

Three cases again; the dependent source 4i stays in all three. Case I has a supermesh around the 8 A source.



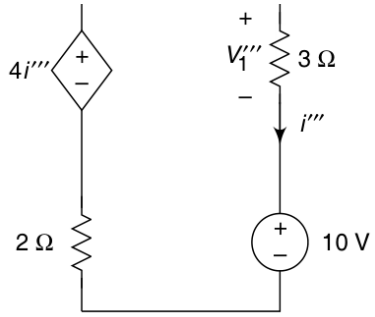
Case I: 8 A alone · Case II: 2 A alone

Case I: 8 A alone. Supermesh KVL and KCL at node x:

$$3i' + 2i_1 - 4i' = 0 \Rightarrow i_1 = \frac{i'}{2}; \qquad i_1 = 8 + i' \Rightarrow \frac{i'}{2} = 8 + i' \Rightarrow i' = -16 \text{ A}, \qquad V_1' = 3i' = -48 \text{ V}$$

Case II: 2 A alone. Mesh current i_2 on the left; the 3 Ω carries $i'' = i_2 + 2$:

$$3(i_2 + 2) + 2i_2 - 4i'' = 0 \Rightarrow 5i_2 + 6 - 4(i_2 + 2) = 0 \Rightarrow i_2 = 2 \text{ A}, \qquad i'' = 4 \text{ A}, \qquad V_1'' = 12 \text{ V}$$



Case III: 10 V alone

Case III: 10 V alone. One loop:

$$3i''' - 10 + 2i''' - 4i''' = 0 \Rightarrow i''' = 10\ \text{A}, \qquad V_1''' = 30\ \text{V}$$

ANSWER

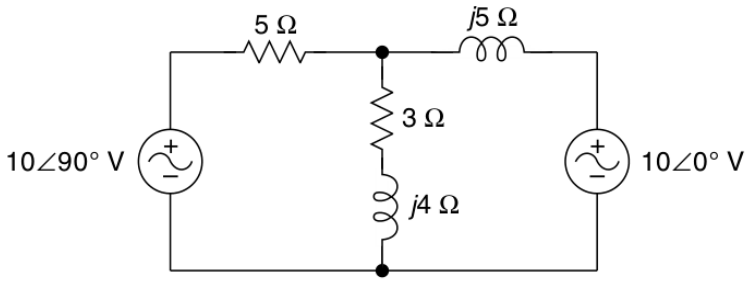
$V_1 = -48 + 12 + 30 = -6$ V (= book)

A12 · Problem 5.12

Superposition · AC **new idea**

QUESTION

Calculate the current through the impedance (3 + j4) Ω using the superposition theorem.

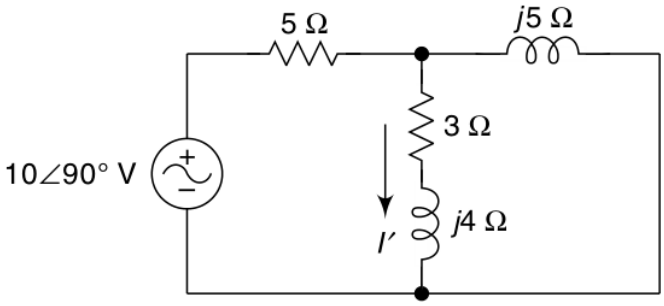


TRY IT FIRST

Two voltage sources, same frequency. Get each case's current as a fraction before simplifying.

HINT: WHAT'S NEW

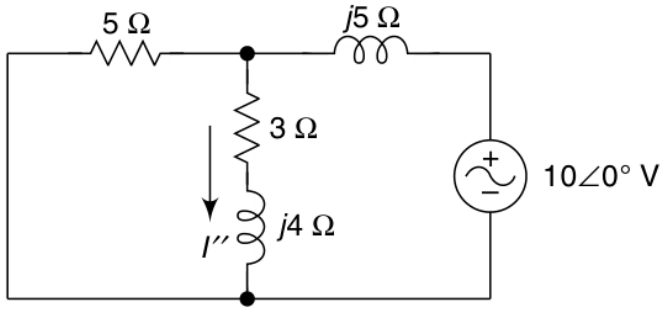
- **The two contributions can cancel.** Here they come out equal and opposite, so the branch current is zero. A zero answer is not an error; check it with one node equation.



10∠90° V alone

10∠90° alone: $I = \dfrac{j10}{5 + \dfrac{(3 + j4)j5}{3 + j9}} = \dfrac{j10(3 + j9)}{-5 + j60}$, $I' = I \times \dfrac{j5}{3 + j9} = \dfrac{-10}{-1 + j12} =$

$0.069 + j0.828$ A



10∠0° V alone

10∠0° alone: $I = \dfrac{10}{j5 + \dfrac{5(3 + j4)}{8 + j4}} = \dfrac{10(8 + j4)}{-5 + j60}$, $I'' = I \times \dfrac{5}{8 + j4} = \dfrac{10}{-1 + j12} =$

$-0.069 - j0.828$ A

ANSWER

I = I′ + I″ = 0 A (= book)

CHECK

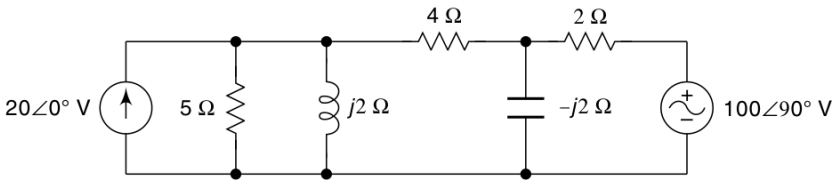
Node equation at the top node: $V(\frac{1}{5} + \frac{1}{3 + j4} + \frac{1}{j5}) = \frac{j10}{5} + \frac{10}{j5} = j2 - j2 = 0$, so V = 0 and no current flows in 3 + j4 ✓.

A13 · Problem 5.13

Superposition · AC **practice**

QUESTION

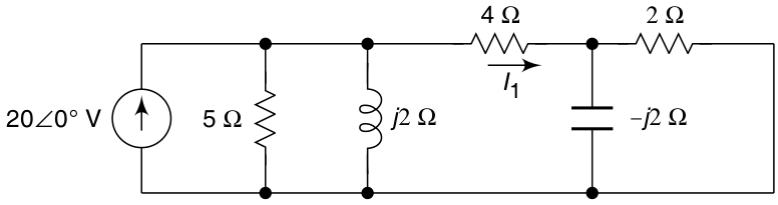
Using the superposition theorem, determine the current in the 4 Ω resistor.



The book labels the current source "20∠0° V"; it is a current source (arrow), 20∠0° A.

TRY IT FIRST

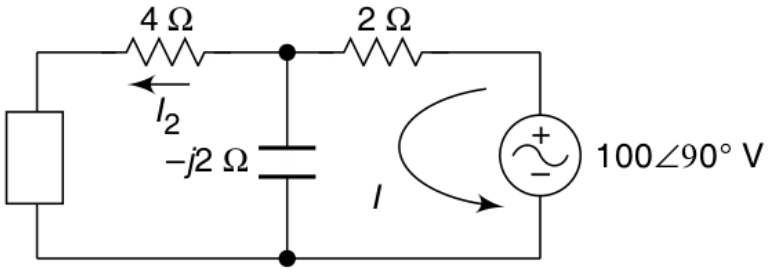
Combine the parallel pairs first: 5 Ω ∥ j2 Ω and 2 Ω ∥ −j2 Ω.



Case I: 20∠0° A alone

Case I: 20∠0° A alone (voltage source shorted). $Z_1 = 5 \parallel j2 = 0.690 + j1.724\ \Omega$; $Z_2 = 2 \parallel (-j2) = 1 - j1\ \Omega$.

$$I_1 = 20 \times \frac{Z_1}{Z_1 + 4 + Z_2} = 3.145 + j5.660\ \text{A} \qquad \text{(left} \rightarrow \text{right)}$$



Case II: 100∠90° V alone

Case II: 100∠90° V alone (current source opened). $(Z_1 + 4) \parallel (-j2) = 0.850 - j1.950\ \Omega$:

$$I_s = \frac{j100}{2 + (0.850 - j1.950)} = -16.35 + j23.90\ \text{A}, \qquad V_B = I_s(0.850 - j1.950) = 32.70 + j52.20\ \text{V}$$

$$I_2 = \frac{V_B}{Z_1 + 4} = 9.748 + j7.547\ \text{A} \qquad \text{(right} \rightarrow \text{left)}$$

ANSWER

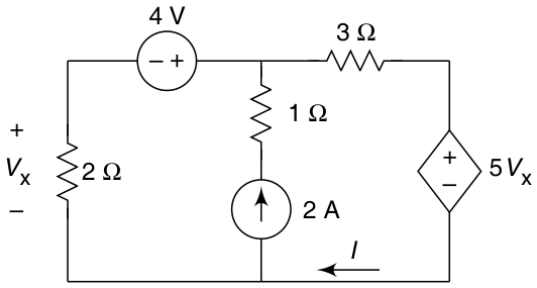
$I = I_1 - I_2 = -6.604 - j1.887 = 6.87\angle -164.05°$ A (left → right), i.e. 6.87 A flowing right → left (= book, 6.89∠−163.67° A, to rounding)

A14 · Problem 5.14

Superposition · dependent source **practice**

QUESTION

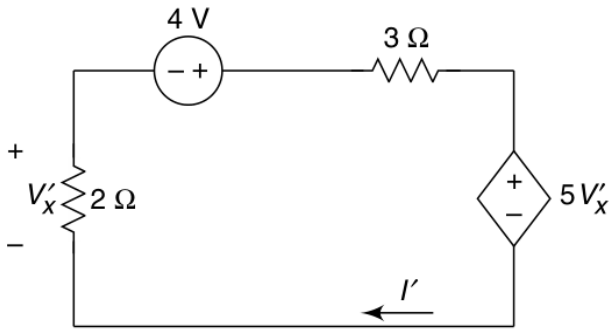
Find I in the figure using the superposition theorem.



$5V_x$ is a voltage-controlled voltage source; V_x is across the 2 Ω (+ at top).

TRY IT FIRST

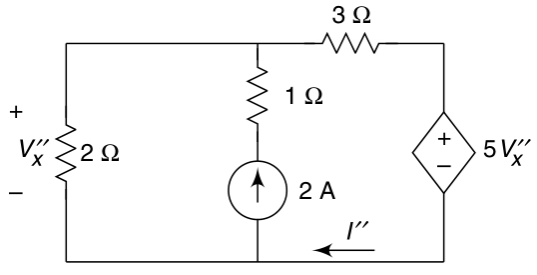
Same idea as 5.3. In case II, the 4 V short joins the two top-left nodes; write one KCL there.



Case I: 4 V alone

Case I: 4 V alone (2 A opened). One loop; I′ flows up the 2 Ω, so $V_x' = -2I'$:

$$-4 + 3I' + 5V_x' - V_x' = 0 \Rightarrow 3I' + 4(-2I') = 4 \Rightarrow I' = -0.8 \text{ A}$$



Case II: 2 A alone

Case II: 2 A alone (4 V shorted). KCL at the top node (voltage V_x''):

$$2 = \frac{V_x''}{2} + \frac{V_x'' - 5V_x''}{3} \Rightarrow V_x'' = -2.4 \text{ V}, \qquad I'' = \frac{V_x'' - 5V_x''}{3} = 3.2 \text{ A}$$

ANSWER

I = −0.8 + 3.2 = 2.4 A (= book)

B · Thevenin's and Norton's theorems

Superposition: one source at a time (done) → **Thevenin: whole network → one source + one impedance** → Norton: the same as a current source → Maximum power transfer (next)

Where this sits. Superposition showed that a linear circuit's response is a sum of simple parts. Thevenin's theorem uses that fact to shrink everything a load is connected to into **two numbers**, so changing the load never means re-solving the whole circuit.

Three terms first.

- **Terminals a–b:** the two points where the load connects. Everything on the other side of them is the **network**.
- **Open-circuit voltage V_{oc}:** the voltage across a–b with the load removed (nothing connected).
- **Short-circuit current I_{sc}:** the current through a wire placed across a–b, counted **from a to b through the wire** (out of the + terminal).

Statement. Seen from a–b, any linear network behaves exactly like **one voltage source $V_{th} = V_{oc}$ in series with one impedance Z_{th}**. Norton's form is the same network as **one current source $I_N = I_{sc}$ in parallel with $Z_N = Z_{th}$**. The two forms are a source transformation of each other: $V_{th} = I_N Z_{th}$.

Finding Z_{th}: three cases.

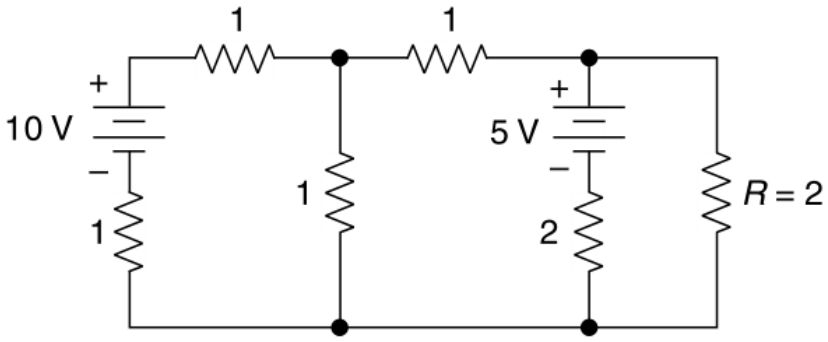
The network contains	Z_{th} is found by
Independent sources only	switch every source off (V → short, I → open), then combine impedances seen from a–b
Independent **and** dependent sources	$Z_{th} = V_{oc} / I_{sc}$, or switch off the independent sources only and apply a test source
Dependent sources only	$V_{th} = 0$; apply a test source (1 A in, measure V): $Z_{th} = V / I$

With dependent sources Z_{th} can come out **negative** (the network supplies energy). Keep the sign of I_{sc}: it decides the sign of Z_{th}.

A15 · Problem 5.15

Thevenin · DC **teach**

> **QUESTION**
> Draw the Thevenin's equivalent of the circuit in the figure and find the load current i. All values are in ohm.



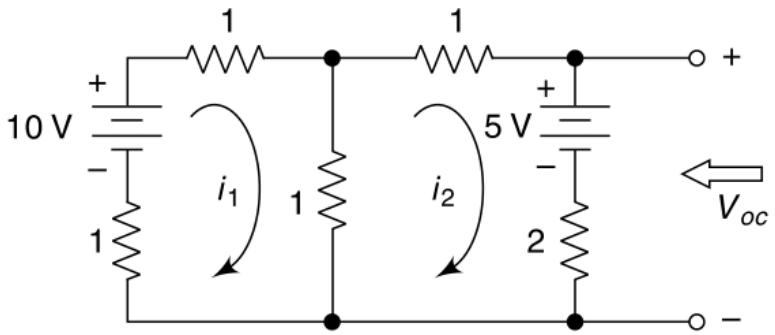
i flows down through R = 2 Ω.

TRY IT FIRST

Remove R. Find the voltage across its terminals (two meshes), then short both sources and find the resistance seen from there. Last, put R back on the equivalent.

HINT: WHAT'S NEW

- **The Thevenin recipe:** (1) remove the load; (2) V_{th} = open-circuit voltage at its terminals; (3) R_{th} = resistance seen into the terminals with every independent source switched off; (4) load current = $V_{th} / (R_{th} + R_L)$.
- **Switching off for R_{th}** uses the same rule as superposition: voltage sources become shorts.



Load removed: two mesh currents, V_{oc} across the open terminals

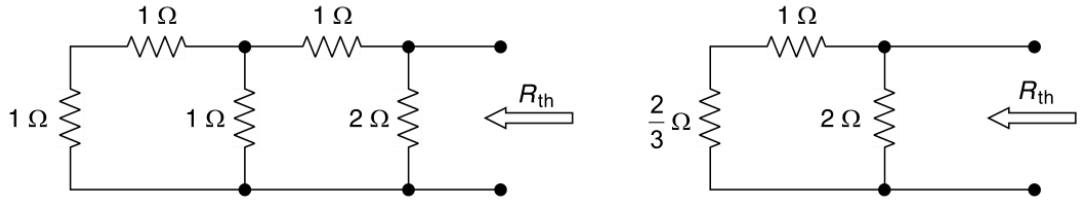
Given: 10 V, 5 V; 1 Ω (series with 10 V), 1 Ω, 1 Ω, 2 Ω; R = 2 Ω. **To find:** V_{th}, R_{th}, i.

Step 1: V_{th} (R removed). Mesh currents clockwise:

$$\text{Mesh 1:} \quad 3i_1 - i_2 = 10 \qquad \text{Mesh 2:} \quad -i_1 + 4i_2 = -5$$

$$\Rightarrow \quad i_1 = \frac{35}{11}\ \text{A}, \qquad i_2 = -\frac{5}{11}\ \text{A}$$

$$V_{th} = V_{oc} = 5 + 2i_2 = 5 - \frac{10}{11} = \frac{45}{11} = 4.091\ \text{V}$$



Both sources shorted: R_{th} seen from the load terminals

Step 2: R_{th} (both sources shorted):

$$(1 + 1) \parallel 1 = \frac{2}{3}\ \Omega, \qquad \frac{2}{3} + 1 = \frac{5}{3}\ \Omega, \qquad R_{th} = \frac{5}{3} \parallel 2 = \frac{10}{11} = 0.909\ \Omega$$

Step 3: load current:

$$i = \frac{V_{th}}{R_{th} + R} = \frac{45/11}{10/11 + 2} = \frac{45}{32}\ A$$

ANSWER

Thevenin equivalent: 45/11 V in series with 10/11 Ω. i = 1.406 A (= book)

SIDE NOTE

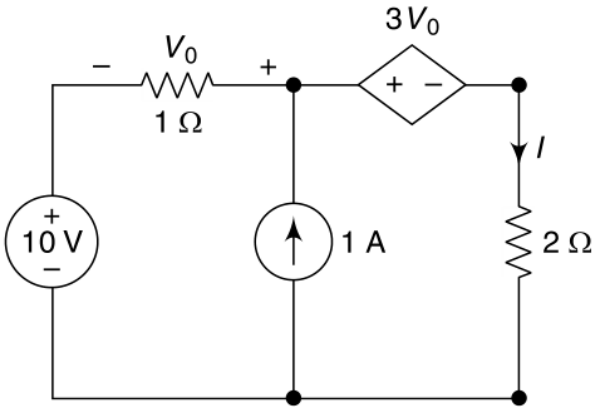
The book prints $i_1 = 5/11$; it is 35/11 (only i_2 is used, so the answer is unaffected). Full node solve with R in place: i = 1.40625 A ✓.

A16 · Problem 5.16

Thevenin · dependent source **teach**

QUESTION

Find I in the given figure, using Thevenin's theorem.



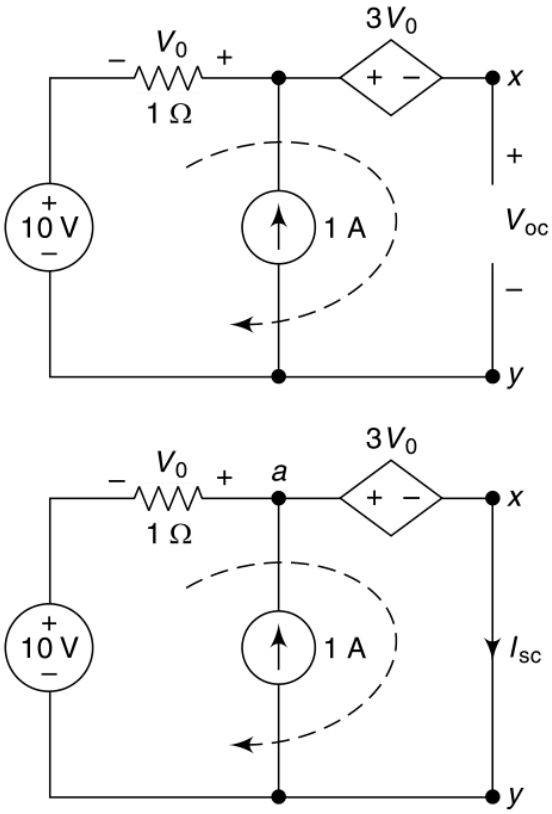
$3V_0$ is a voltage-controlled voltage source; V_0 is across the 1 Ω (+ on the right).

TRY IT FIRST

The dependent source cannot be switched off, so R_{th} must come from V_{oc} / I_{sc}. Find both, and keep the sign of I_{sc}.

HINT: WHAT'S NEW

- **Dependent source ⇒ $R_{th} = V_{oc} / I_{sc}$** (or a test source). Shorting the independent sources and combining resistors would ignore the dependent source.
- **I_{sc} is measured from x to y through the short.** A negative I_{sc} gives a **negative R_{th}**, which is allowed: the dependent source can supply energy.



Top: terminals x–y open (V_{oc}). Bottom: x–y shorted (I_{sc}).

Given: 10 V, 1 A, 1 Ω, VCVS $3V_0$, load 2 Ω. **To find:** I.

V_{oc} (2 Ω removed). Nothing leaves at x, so the 1 A must flow left through the 1 Ω: $V_0 = 1 \times 1 = 1$ V.

$$V_a = 10 + V_0 = 11 \text{ V}, \qquad V_{oc} = V_a - 3V_0 = 11 - 3 = 8 \text{ V}$$

I_{sc} (x shorted to y, so $V_x = 0$):

$$0 = V_a - 3V_0 = (10 + V_0) - 3V_0 \qquad \Rightarrow \qquad V_0 = 5 \text{ V}$$

$$\text{KCL at a:} \qquad 1 = \frac{V_0}{1} + I_{sc} \qquad \Rightarrow \qquad I_{sc} = 1 - 5 = -4\ \text{A}$$

$$R_{th} = \frac{V_{oc}}{I_{sc}} = \frac{8}{-4} = -2\ \Omega$$

Load: $R_{th} + 2\ \Omega = -2 + 2 = 0$, so

$$I = \frac{V_{th}}{R_{th} + 2} = \frac{8}{0} \qquad \Rightarrow \qquad \text{no finite solution}$$

ANSWER

$V_{th} = 8$ V, $R_{th} = -2\ \Omega$. With the 2 Ω load, $R_{th} + R_L = 0$: the circuit as drawn has no solution. (Book: 2 A.)

BOOK ERROR IN 5.16

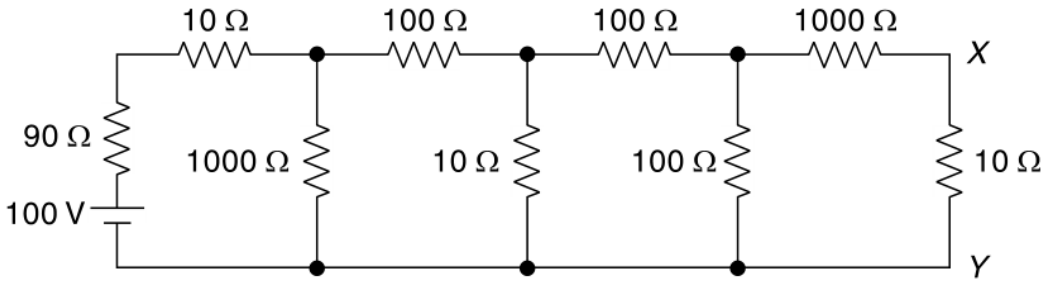
The book finds $I_{sc} = -4$ A, then writes $R_{th} = 8/4 = 2\ \Omega$, dropping the minus sign, and prints $I = 2$ A. Direct check with the 2 Ω in place: KCL at a gives $V_0 = 1 - I$; the right-hand path gives $V_x = 10 - 2V_0 = 8 + 2I$; but $V_x = 2I$, so $8 = 0$, a contradiction. A test source (10 V shorted, 1 A opened, 1 A pushed into x) confirms $R_{th} = -2\ \Omega$. For any other load, $I = \dfrac{8}{R_L - 2}$.

A17 · Problem 5.17

Thevenin · repeated reduction **new idea**

QUESTION

By the iterative use of Thevenin's theorem, reduce the circuit shown in the figure to a single emf acting in series with a single resistor. Hence, calculate the current in the 10 Ω resistor connected across XY.



Ladder network; the load is the 10 Ω across XY.

TRY IT FIRST

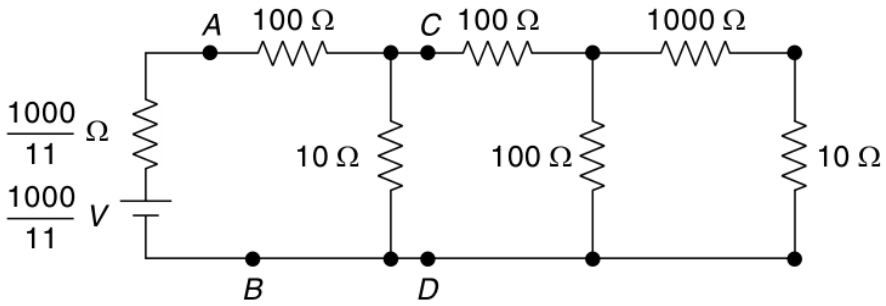
Cut the ladder just right of the first shunt resistor and replace the left part by its Thevenin equivalent. Repeat, one rung at a time.

HINT: WHAT'S NEW

- **Iterative Thevenin:** a ladder is reduced rung by rung. Each step is a voltage divider for V_{th} and a parallel combination for R_{th}: $V_{new} = V_{old} \times \dfrac{R_{sh}}{R_{old} + R_{series} + R_{sh}}$, $R_{new} = (R_{old} + R_{series}) \parallel R_{sh}$.

Step 1 (left of AB: 100 V behind 90 + 10 = 100 Ω, shunt 1000 Ω):

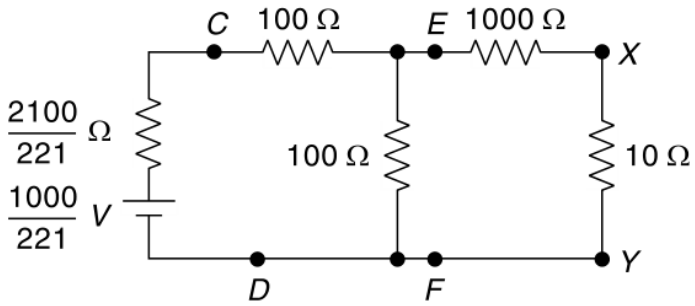
$$V_1 = 100 \times \frac{1000}{1100} = \frac{1000}{11}\ \text{V}, \qquad R_1 = 100 \parallel 1000 = \frac{1000}{11}\ \Omega$$



After step 1

Step 2 (add 100 Ω series, shunt 10 Ω):

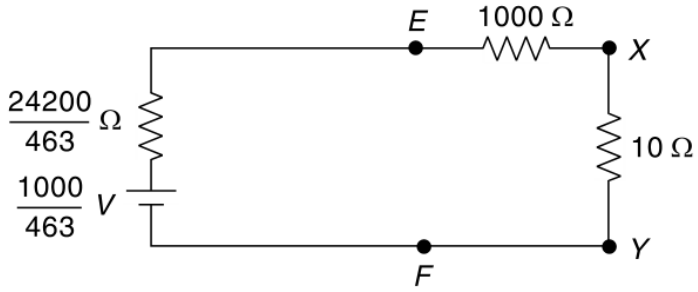
$$V_2 = \frac{1000}{11} \times \frac{10}{2100/11 + 10} = \frac{1000}{221}\ \text{V}, \qquad R_2 = \frac{2100}{11} \parallel 10 = \frac{2100}{221}\ \Omega$$



After step 2

Step 3 (add 100 Ω series, shunt 100 Ω):

$$V_3 = \frac{1000}{221} \times \frac{100}{24200/221 + 100} = \frac{1000}{463}\ \text{V}, \qquad R_3 = \frac{24200}{221} \parallel 100 = \frac{24200}{463}\ \Omega$$



After step 3, with the 1000 Ω still in series

Step 4 (add 1000 Ω series):

$$V_{th} = \frac{1000}{463} = 2.160\ \text{V}, \qquad R_{th} = \frac{24200}{463} + 1000 = \frac{487200}{463} = 1052.3\ \Omega$$

$$I = \frac{1000/463}{487200/463 + 10} = \frac{1000}{491830}\ \text{A}$$

ANSWER

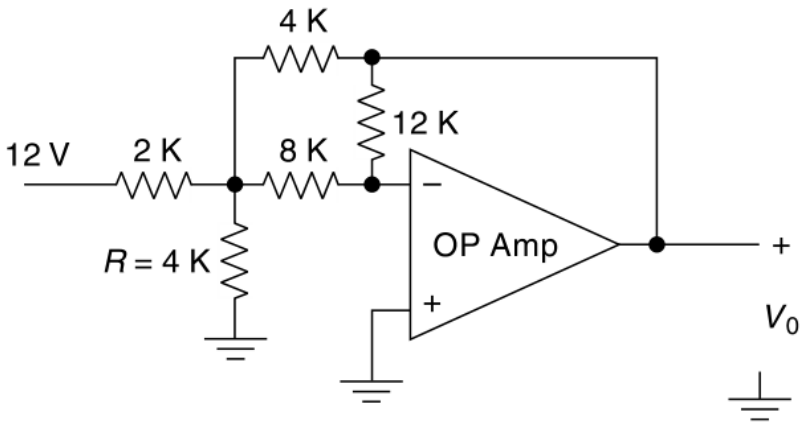
Single emf 2.160 V in series with 1052.3 Ω; I = 2.03 mA ≈ 0.002 A (= book)

A18 · Problem 5.18

Thevenin · op-amp, test source **new idea**

QUESTION

In the operational-amplifier circuit shown in the figure, find I in the R = 4 kΩ resistor, using Thevenin's theorem.



Ideal op-amp; the load is R = 4 kΩ from node e_1 to ground.

TRY IT FIRST

With R removed, write KCL at e_1 and at the inverting input. For R_{th}, short the 12 V and push 1 A into e_1.

HINT: WHAT'S NEW

- **Ideal op-amp rules:** no current into its inputs, and (with negative feedback) $e_2 = e_+ = 0$ V. The output can supply any current, so **no KCL is written at the output node**.
- **Test source for R_{th}:** the op-amp acts like a dependent source, so switch off the 12 V only, inject 1 A at the terminals and read the voltage: $R_{th} = e_1 / 1$ A.

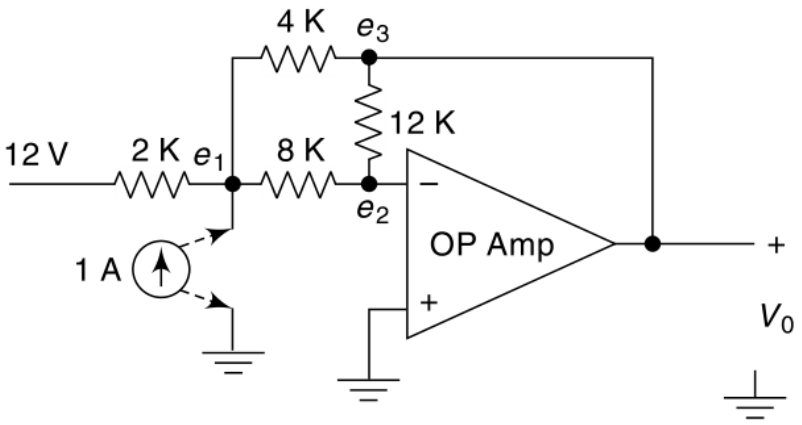
Given: 12 V, 2 kΩ, 4 kΩ (feedback $e_1 \to V_0$), 8 kΩ, 12 kΩ, ideal op-amp. **To find:** I in R = 4 kΩ.

V_{th} (R removed; $e_2 = 0$, $e_3 = V_0$):

$$\text{KCL at } e_2: \quad \frac{0 - e_1}{8k} + \frac{0 - V_0}{12k} = 0 \quad \Rightarrow \quad V_0 = -\frac{3}{2}e_1$$

$$\text{KCL at } e_1: \quad \frac{e_1 - 12}{2k} + \frac{e_1 - V_0}{4k} + \frac{e_1}{8k} = 0 \quad \Rightarrow \quad 7e_1 = 48 + 2V_0$$

$$7e_1 = 48 - 3e_1 \quad \Rightarrow \quad V_{th} = e_1 = 4.8 \text{ V}$$



12 V shorted, 1 A test source at the load terminals

R_{th} (12 V shorted, 1 A injected at e_1):

$$\frac{e_1}{2k} + \frac{e_1 - V_0}{4k} + \frac{e_1}{8k} = 1 \quad \Rightarrow \quad 7e_1 - 2V_0 = 8000$$

$$10e_1 = 8000 \quad \Rightarrow \quad e_1 = 800 \text{ V}$$

$$R_{th} = \frac{800 \text{ V}}{1 \text{ A}} = 800 \ \Omega, \qquad I = \frac{4.8}{800 + 4000} = 1 \times 10^{-3} \text{ A}$$

ANSWER

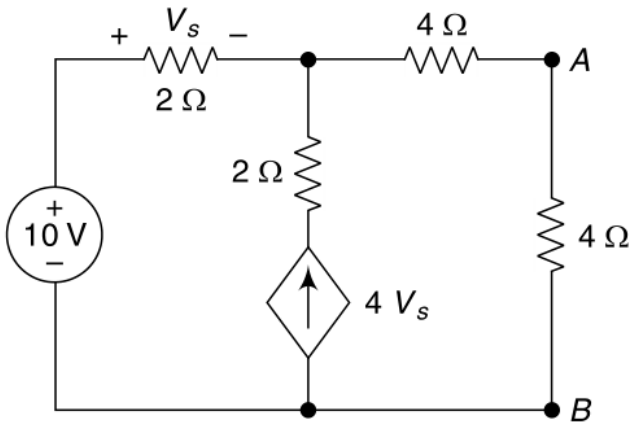
V_{th} = 4.8 V, R_{th} = 800 Ω, I = 1 mA (= book)

A19 · Problem 5.19

Thevenin · dependent current source **practice**

QUESTION

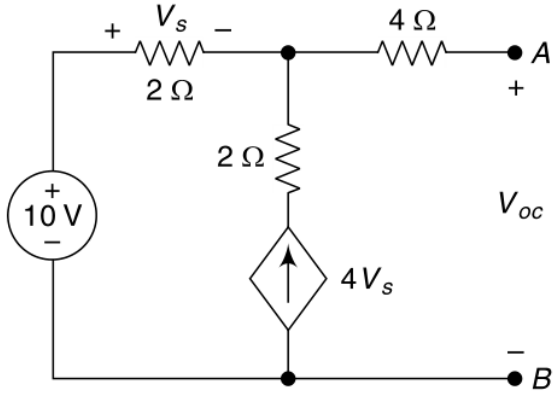
Find Thevenin's equivalent about AB for the circuit shown in the figure.



The 4 Ω across AB is the load (the book removes it).

TRY IT FIRST

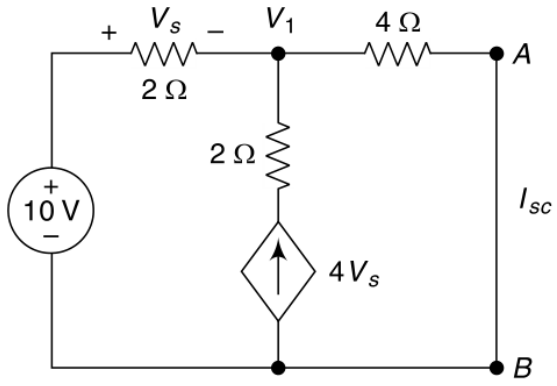
V_{oc}: with AB open, what does KCL at the middle node say about V_s? Then short AB for I_{sc}.



AB open

V_{oc} (AB open, so the series 4 Ω carries nothing; $V_s = 10 - V_M$). KCL at the middle node M:

$$\frac{V_s}{2} + 4V_s = 0 \quad \Rightarrow \quad V_s = 0 \quad \Rightarrow \quad V_{oc} = V_M = 10\ \text{V}$$



AB shorted

I_{sc} (AB shorted; $V_1 = V_M$, $V_s = 10 - V_1$):

$$\frac{V_1 - 10}{2} + \frac{V_1}{4} = 4(10 - V_1) \quad \Rightarrow \quad 19V_1 = 180 \quad \Rightarrow \quad V_1 = 9.474 \text{ V}$$

$$I_{sc} = \frac{V_1}{4} = 2.368 \text{ A}$$

$$R_{th} = \frac{10}{2.368} = 4.22 \ \Omega$$

ANSWER

V_{th} = 10 V, R_{th} = 38/9 = 4.22 Ω (= book)

SIDE NOTE

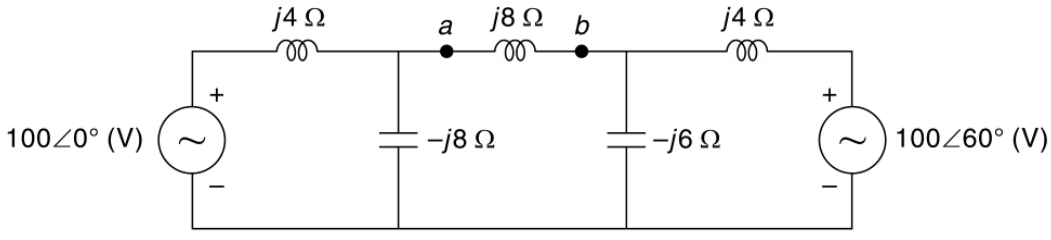
The 2 Ω in series with the dependent current source never matters (5.1's shortcut). Test-source check (10 V shorted, 1 A into A): $V_A = 4 + \frac{1}{4.5} = 4.22$ V ⇒ $R_{th} = 4.22$ Ω ✓.

A20 · Problem 5.20

Thevenin · AC, floating terminals **new idea**

QUESTION

In the network, determine the steady current in the 8 Ω inductor using Thevenin's theorem.



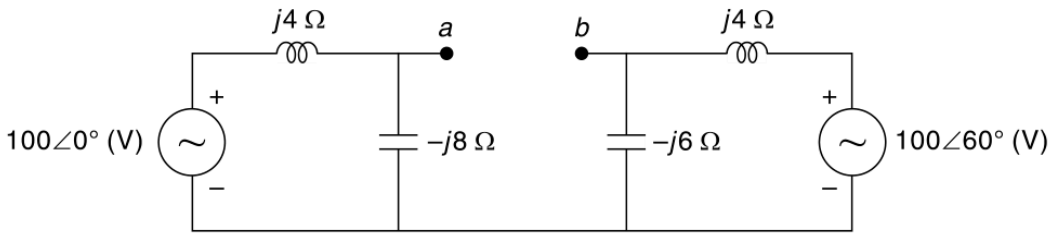
The load is the j8 Ω between a and b.

TRY IT FIRST

With j8 removed, the circuit splits into two separate halves. Find V_a and V_b with a divider each.

HINT: WHAT'S NEW

- **Neither terminal is ground.** $V_{th} = V_a - V_b$, each found from its own half.
- With the sources shorted, each half is a parallel pair, and the two pairs are **in series** between a and b.



j8 Ω removed: two independent halves

$$V_a = 100\angle 0° \times \frac{-j8}{j4 - j8} = 200\angle 0° \text{ V}, \qquad V_b = 100\angle 60° \times \frac{-j6}{j4 - j6} = 300\angle 60° \text{ V}$$

$$V_{th} = V_a - V_b = 200 - (150 + j259.81) = 50 - j259.81 = 264.58\angle -79.11° \text{ V}$$

$$Z_{th} = \frac{(j4)(-j8)}{j4 - j8} + \frac{(j4)(-j6)}{j4 - j6} = j8 + j12 = j20\ \Omega$$

$$i = \frac{V_{th}}{Z_{th} + j8} = \frac{264.58\angle -79.11°}{28\angle 90°}$$

ANSWER

i = 9.45∠−169.11° A (a → b) (= book)

A21 · Problem 5.21

Thevenin / Norton · AC, s-domain, dependent source **new idea**

QUESTION

Obtain Thevenin's equivalent circuit with respect to terminals A–B in the networks shown below.

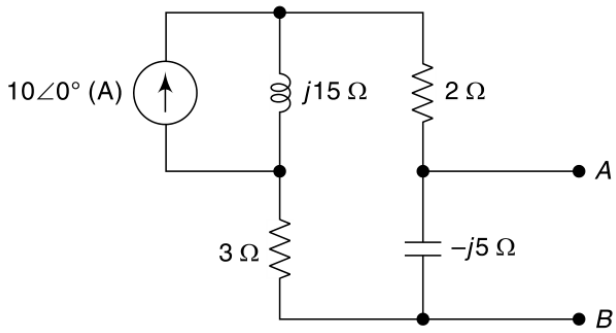
TRY IT FIRST

(a)–(c) are practice. In (d), treat s like jω. In (e) the dependent source stays, so use V_{oc} / I_{sc}.

HINT: WHAT'S NEW

- **(d) s-domain:** impedances are R, sL and $\frac{1}{sC}$; the source becomes its Laplace transform, 2 cos 2t $\rightarrow \frac{2s}{s^2 + 4}$. Everything else is unchanged.
- **(e) AC with a dependent source:** find V_{th} and I_N separately; $Z_{th} = V_{th} / I_N$.

(a)



Source transformation: $10\angle 0°$ A ‖ j15 Ω → j150 V in series with j15 Ω. With AB open, one loop:

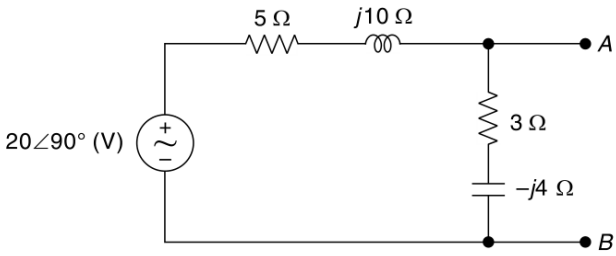
$$I = \frac{j150}{j15 + 2 - j5 + 3} = \frac{j150}{5 + j10} = 13.42\angle 26.57° \text{ A}, \qquad V_{th} = I \times (-j5) = 30 - j60 \text{ V}$$

$$Z_{th} = (-j5) \parallel (5 + j15) = \frac{-j5(5 + j15)}{5 + j10} = 1 - j7 \ \Omega$$

ANSWER

(a) $V_{th} = 67.08\angle -63.43°$ V, $Z_{th} = 7.07\angle -81.87° \ \Omega$ (= book)

(b)



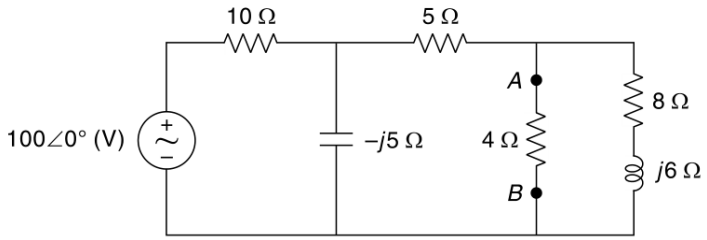
$$V_{th} = 20\angle 90° \times \frac{3 - j4}{(5 + j10) + (3 - j4)} = \frac{j20(3 - j4)}{8 + j6} = \frac{100\angle 36.87°}{10\angle 36.87°} = 10\angle 0° \text{ V}$$

$$Z_{th} = \frac{(5 + j10)(3 - j4)}{8 + j6} = \frac{55 + j10}{8 + j6} = 5 - j2.5 \ \Omega$$

ANSWER

(b) $V_{th} = 10\angle 0°$ V, $Z_{th} = 5.59\angle -26.57°$ Ω (= book; its first line prints j120, a typo for j20)

(c) The 4 Ω across AB is the load.



$$Z = 10 + (-j5) \parallel (13 + j6) = 11.91 - j5.15 = 12.98\angle -23.37° \ \Omega$$

$$I = \frac{100}{Z} = 7.706\angle 23.37° \text{ A}$$

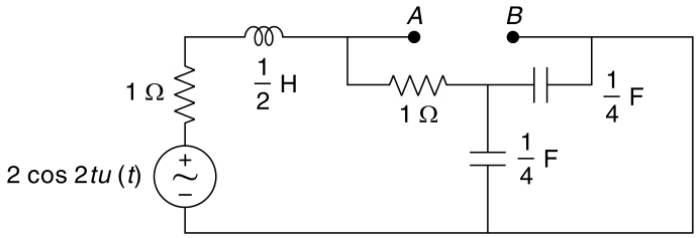
$$I_{right} = I \times \frac{-j5}{13 + j1} = 2.955\angle -71.03° \text{ A}, \qquad V_{th} = I_{right}(8 + j6) = 29.55\angle -34.16° \text{ V}$$

$$Z_{th} = [(10 \parallel -j5) + 5] \parallel (8 + j6) = (7 - j4) \parallel (8 + j6) = 5.328 - j0.044 \ \Omega$$

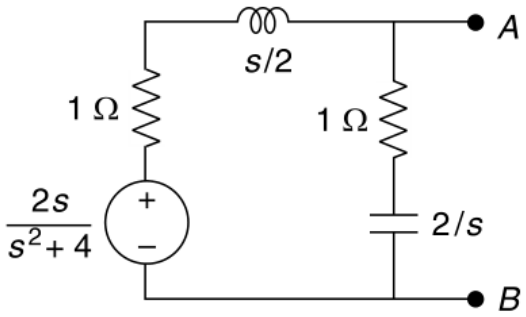
ANSWER

(c) $V_{th} = 29.55\angle -34.16°$ V, $Z_{th} = 5.33\angle -0.47°$ Ω (= book)

(d)



The two $\frac{1}{4}$ F capacitors both run from the same node to ground, so they add: $\frac{1}{2}$ F → $\frac{2}{s}$. A is the node after the inductor; B is ground.



Redrawn in the s-domain

$$V_{th}(s) = \frac{2s}{s^2+4} \times \frac{1 + 2/s}{(1 + 2/s) + (1 + s/2)} = \frac{2s}{s^2+4} \times \frac{2}{s+2} = \frac{4s}{(s^2+4)(s+2)}$$

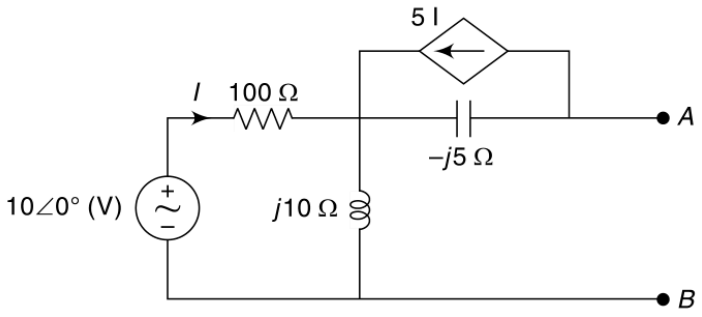
(Simplify: $1 + 2/s = \frac{s+2}{s}$ and $2 + s/2 + 2/s = \frac{(s+2)^2}{2s}$, so the ratio is $\frac{2}{s+2}$.)

$$Z_{th}(s) = (1 + 2/s) \parallel (1 + s/2) = \frac{(s+2)^2/(2s)}{(s+2)^2/(2s)} = 1\ \Omega$$

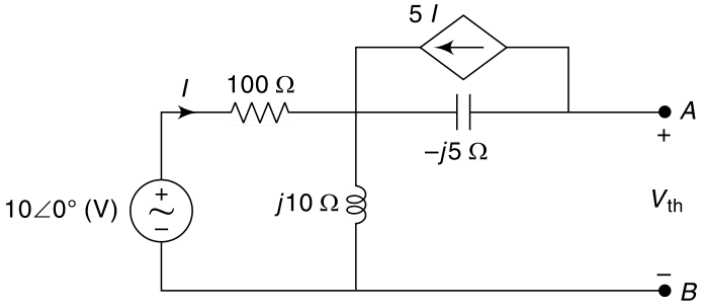
ANSWER

(d) $V_{th}(s) = \frac{4s}{(s^2+4)(s+2)}$ V, $Z_{th} = 1\ \Omega$ (= book)

(e)



I flows through the 100 Ω; the dependent source 5I points from A towards the middle node.

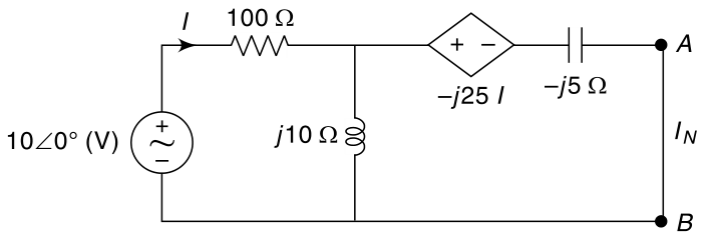


AB open

V_{th}. With AB open, the 5I circulates only through the −j5 Ω, so the source current is set by the left loop alone:

$$I = \frac{10\angle 0°}{100 + j10} = 0.0995\angle -5.71° \text{ A}$$

$$V_{th} = V_{AB} = I(j10) - 5I(-j5) = j35I = 3.483\angle 84.29° \text{ V}$$



AB shorted, with 5I ‖ −j5 transformed to −j25I in series with −j5

I_N. AB shorted. Right loop: $j10(I - I_N) + j25I + j5I_N = 0 \Rightarrow I_N = 7I$. Left loop:

$$10 = (100 + j10)I - j10I_N = (100 - j60)I \qquad \Rightarrow \qquad I = 0.0857\angle 30.96° \text{ A}$$

$$I_N = 7I = 0.600\angle 30.96° \text{ A}$$

$$Z_{th} = \frac{V_{th}}{I_N} = \frac{3.483\angle 84.29°}{0.600\angle 30.96°} = 5.80\angle 53.33° \ \Omega$$

ANSWER

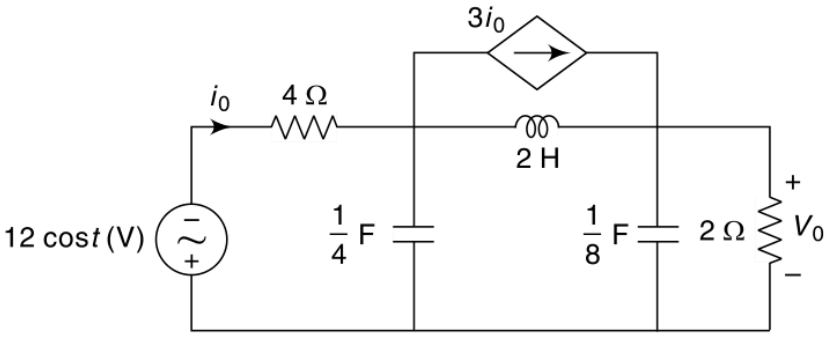
(e) $V_{th} = 3.48\angle 84.3°$ V, $I_N = 0.600\angle 31.0°$ A, $Z_{th} = 5.80\angle 53.3°$ Ω (= book)

A22 · Problem 5.22

Thevenin · AC, dependent source **practice**

QUESTION

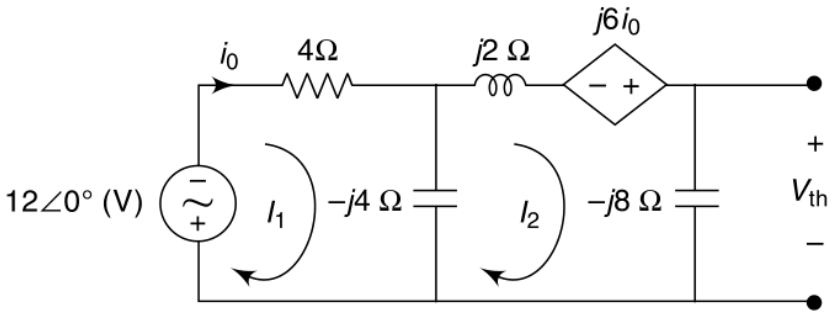
Find V_0 using Thevenin's theorem.



ω = 1 rad/s. The source's + is at the bottom; i_0 flows right through the 4 Ω.

TRY IT FIRST

$\omega = 1$: work out the three reactances first. Transform $3i_0 \parallel j2$ into a voltage source, then write two mesh equations for each of V_{th} and I_N.



2 Ω removed; $3i_0 \parallel j2\ \Omega \rightarrow j6i_0$ in series with j2 Ω (+ on the right)

Reactances ($\omega = 1$): 2 H → j2 Ω, $\frac{1}{4}$ F → −j4 Ω, $\frac{1}{8}$ F → −j8 Ω.

V_{th} (2 Ω removed; clockwise mesh currents, $i_0 = I_1$):

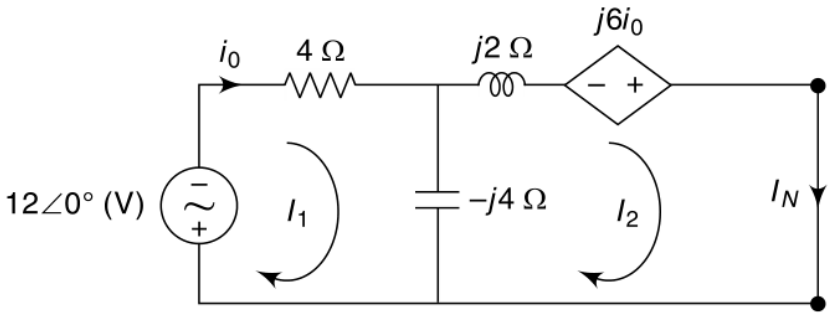
$$\text{Mesh 1:} \qquad (4 - j4)I_1 + j4I_2 = -12$$

$$\text{Mesh 2:} \qquad -j4(I_2 - I_1) + j2I_2 - j6I_1 - j8I_2 = 0 \quad \Rightarrow \quad -j2I_1 - j10I_2 = 0$$

$$\Rightarrow \quad I_2 = -0.2I_1$$

$$(4 - j4.8)I_1 = -12 \quad \Rightarrow \quad I_1 = -1.230 - j1.475\ \text{A}, \qquad I_2 = 0.246 + j0.295\ \text{A}$$

$$V_{th} = I_2(-j8) = 2.361 - j1.967 = 3.073\angle -39.81°\ \text{V}$$



Terminals shorted (the −j8 is shorted out)

I_N (terminals shorted):

$$\text{Mesh 2:} \qquad -j4(I_2 - I_1) + j2I_2 - j6I_1 = 0 \quad \Rightarrow \quad I_2 = -I_1; \qquad (4 - j8)I_1 = -12$$

$$I_N = I_2 = \frac{12}{4 - j8} = 0.6 + j1.2 = 1.342\angle 63.43^\circ \text{ A}$$

$$Z_{th} = \frac{V_{th}}{I_N} = 2.290\angle -103.24^\circ = -0.525 - j2.230\ \Omega$$

$$V_0 = V_{th} \times \frac{2}{Z_{th} + 2} = \frac{3.073\angle -39.81^\circ \times 2}{2.674\angle -56.50^\circ} = 2.299\angle 16.70^\circ \text{ V}$$

ANSWER

$V_0 = 2.30\angle 16.70^\circ$ V, i.e. $v_0(t) = 2.30\cos(t + 16.70^\circ)$ V

BOOK ERROR IN 5.22

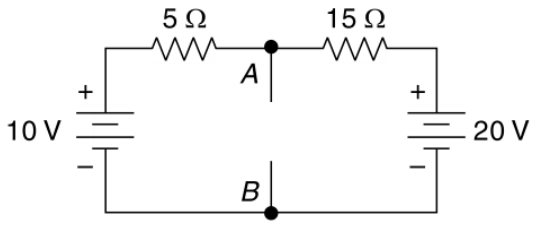
For V_{th}, the book writes mesh 2 as $-j2I_1 - j6I_2 = 0$: it keeps the shared $-j4$ in the I_1 term but drops it from I_2's own impedance ($-j4 + j2 - j8 = -j10$, not $-j6$). That gives $V_{th} = 4.8\angle -36.87^\circ$ V, $Z_{th} = 3.58\angle -100.3^\circ\ \Omega$ and $V_0 = 1.27\angle 32^\circ$ V. Its $I_N = 1.342\angle 63.43^\circ$ A is right. The answer above was confirmed by a full node solve with the 2 Ω in place ($2.2988\angle 16.70^\circ$ V).

A23 · Problem 5.23

Norton · DC **new idea**

QUESTION

Obtain the Norton's equivalent circuit with respect to the terminals AB for the network shown in the figure.

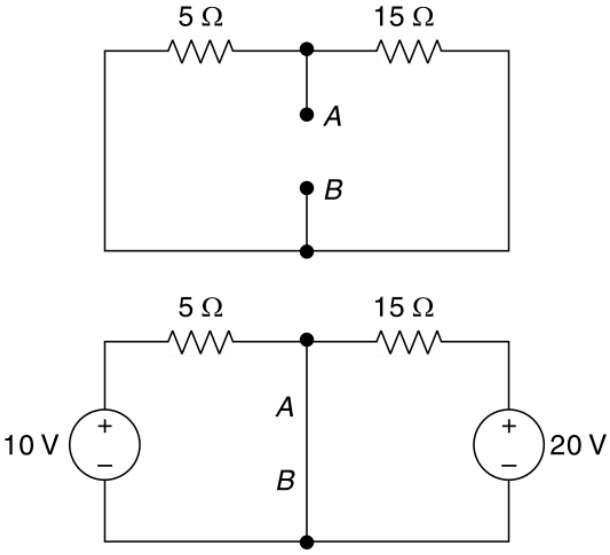


TRY IT FIRST

Short A to B. What current does each source push through the short?

HINT: WHAT'S NEW

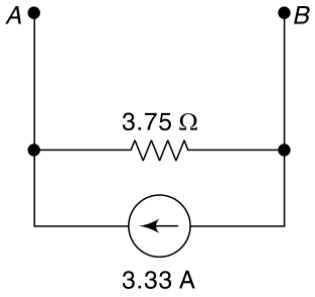
- **Norton's form:** $I_N = I_{sc}$ (current through the A–B short), $R_N = R_{th}$, drawn as a current source in parallel with R_N, arrow pointing towards A.
- With A–B shorted, each source sees only its own resistor, so I_{sc} is a sum of two Ohm's-law currents.



Top: sources removed (R_N). Bottom: A–B shorted (I_N).

$$R_N = 5 \parallel 15 = \frac{75}{20} = 3.75\ \Omega$$

$$I_N = I_{sc} = \frac{10}{5} + \frac{20}{15} = 2 + 1.333 = 3.33\ \text{A}$$



Norton equivalent (arrow towards A)

ANSWER

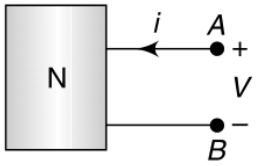
I_N = 3.33 A (towards A) in parallel with R_N = 3.75 Ω (= book). Check: $V_{th} = I_N R_N$ = 12.5 V = node voltage at A ✓.

A24 · Problem 5.24

Norton · from a v–i characteristic **new idea**

QUESTION

For the one-port shown in the figure, determine the Norton's equivalent at the terminals AB, if the v–i characteristic is given by 16v = 80 − 2i.



TRY IT FIRST

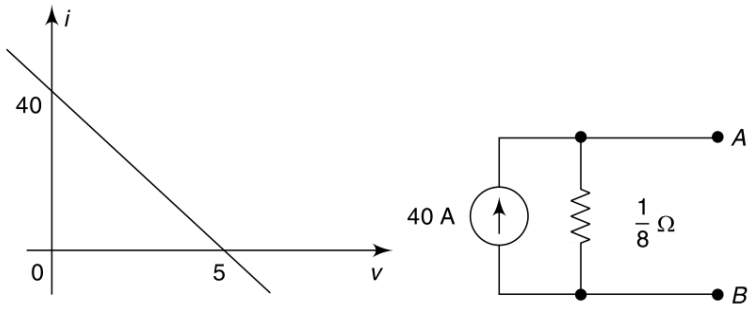
Which two points on the v–i line are V_{oc} and I_{sc}?

HINT: WHAT'S NEW

- **The equivalent is read off the terminal line.** i = 0 gives V_{oc}; v = 0 gives I_{sc}; $R_N = V_{oc} / I_{sc}$ (the size of the line's slope, Δv/Δi).

$$16v = 80 - 2i \qquad \Rightarrow \qquad \frac{v}{5} + \frac{i}{40} = 1$$

$$i = 0: \quad V_{oc} = 5\ V \qquad v = 0: \quad I_{sc} = 40\ A \qquad R_N = \frac{5}{40} = \frac{1}{8}\ \Omega$$



The v–i line and the Norton equivalent

ANSWER

I_N = 40 A (towards A) in parallel with R_N = 1/8 Ω (= book)

SIGN CONVENTION

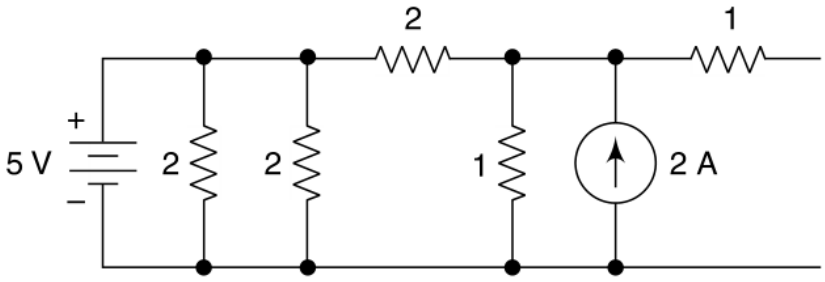
$v = 5 - \frac{i}{8}$ describes an ordinary source only if i is the current **delivered out of A** (v falls as the load draws more). The figure's arrow points into N; read literally, that would make $R_N = -1/8\ \Omega$ with the 40 A reversed. The intended answer is the book's.

A25 · Problem 5.25

Thevenin + Norton · superposition inside **practice**

QUESTION

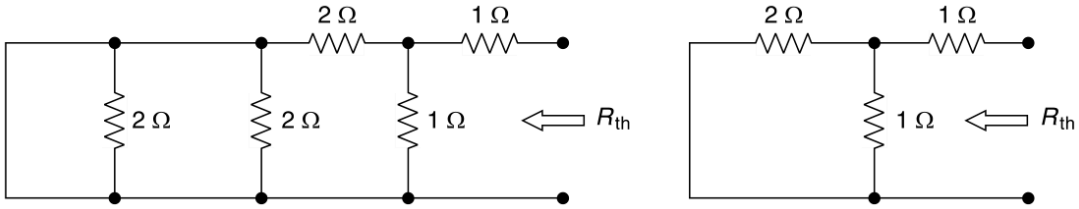
Find both Thevenin's and Norton's equivalent circuits for the network shown in the figure. All values are in ohm.



Terminals at the right-hand end of the 1 Ω series resistor and the bottom rail.

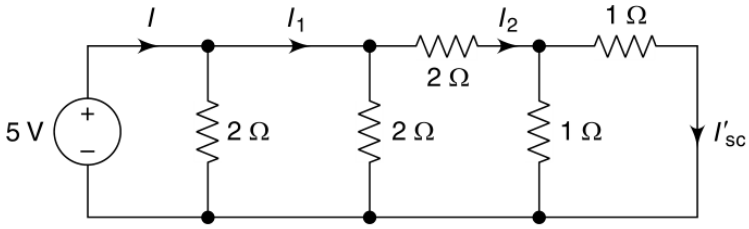
TRY IT FIRST

Which resistors does the shorted 5 V source remove from R_{th}? Then find I_{sc} one source at a time.



Sources switched off: the two 2 Ω across the 5 V are shorted out

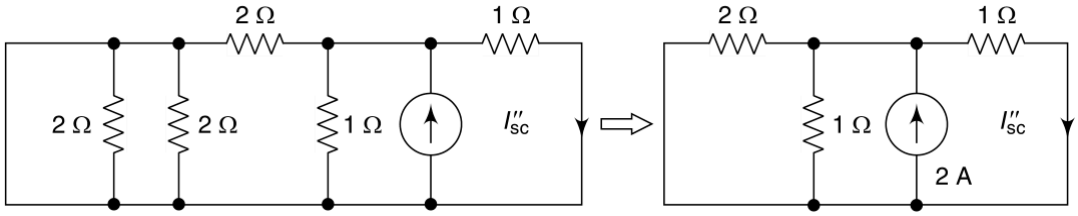
$$R_N = R_{th} = (2 \parallel 1) + 1 = \frac{2}{3} + 1 = \frac{5}{3}\ \Omega$$



5 V alone, terminals shorted

I_{sc}, 5 V alone (2 A opened): the middle node sees 2 Ω to the 5 V, and 1 Ω ∥ 1 Ω to ground:

$$\frac{V_m - 5}{2} + \frac{V_m}{1} + \frac{V_m}{1} = 0 \quad \Rightarrow \quad V_m = 1\ \text{V}, \qquad I_{sc}' = \frac{1}{1} = 1\ \text{A}$$

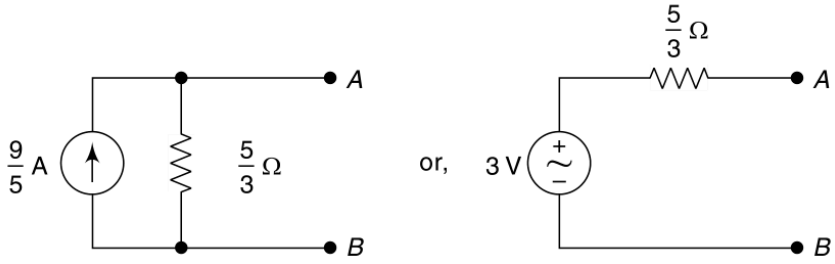


2 A alone, terminals shorted

I_{sc}, 2 A alone (5 V shorted): the 2 A divides between 2 Ω ‖ 1 Ω = 2/3 Ω and the 1 Ω to the short:

$$I_{sc}'' = 2 \times \frac{2/3}{2/3 + 1} = \frac{4}{5}\ \text{A}$$

$$I_N = 1 + \frac{4}{5} = \frac{9}{5}\ \text{A}, \qquad V_{th} = I_N R_{th} = \frac{9}{5} \times \frac{5}{3} = 3\ \text{V}$$



Norton and Thevenin forms

ANSWER

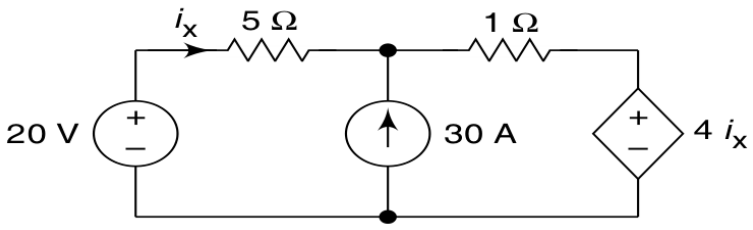
Norton: 1.8 A ‖ 5/3 Ω. Thevenin: 3 V in series with 5/3 Ω (= book)

A26 · Problem 5.26

Thevenin · dependent source **practice**

QUESTION

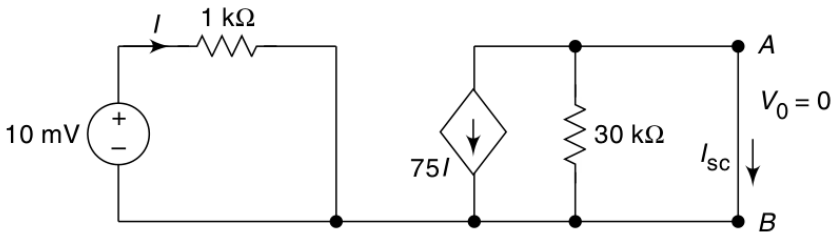
Replace the circuit in the figure with the Thevenin's equivalent circuit across A and B.



The question's figure, as printed: it marks no terminals A and B.

BOOK ERROR IN 5.26: FIGURE AND SOLUTION BELONG TO DIFFERENT CIRCUITS

The figure shows no A and B, so the question cannot be answered as printed. The book's solution solves a different circuit (a transistor model): 10 mV source, 1 kΩ carrying I, a controlled voltage $10^{-4}V_0$ in the input loop, and a controlled current source 75I in parallel with 30 kΩ across A–B. That solution is checked below. Ask your professor which circuit is meant before submitting.



The book's solution circuit, A–B shorted

V_{oc} (A–B open). The 75I pulls current down out of node A, and it can only return through the 30 kΩ, so $V_0 = 30k \times (-75I) = -2.25 \times 10^6 I$. Input loop:

$$10^3 I + 10^{-4}V_0 = 10 \times 10^{-3} \quad \Rightarrow \quad -4.444 \times 10^{-4}V_0 + 1 \times 10^{-4}V_0 = 10^{-2}$$
$$\Rightarrow \quad V_0 = -29.03 \text{ V}$$

I_{sc}. A–B shorted $\Rightarrow V_0 = 0$, so $I = \dfrac{10 \text{ mV}}{1 \text{ k}\Omega} = 10^{-5}$ A and $I_{sc} = -75I = -7.5 \times 10^{-4}$ A.

$$R_{th} = \frac{-29.03}{-7.5 \times 10^{-4}} = 38.71 \text{ k}\Omega$$

ANSWER

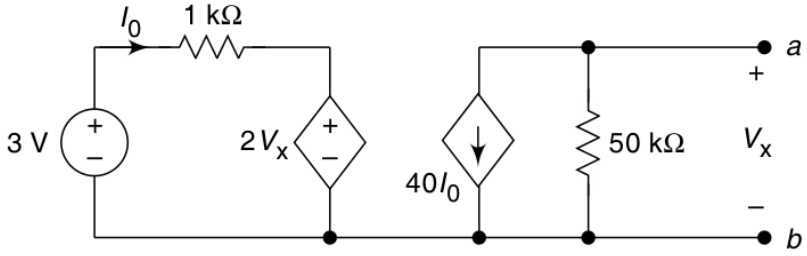
(solution circuit) $V_{th} = -29.0$ V, $R_{th} = 38.7$ kΩ (≈ book: it rounds V_0 to −29 V and prints 38.67 kΩ)

A27 · Problem 5.27

Thevenin · dependent sources, negative R_{th} **practice**

QUESTION

Find the Thevenin's equivalent between terminals a and b of the circuit shown in the figure.



The question figure gives 50 kΩ; the book's solution figure uses 50 Ω.

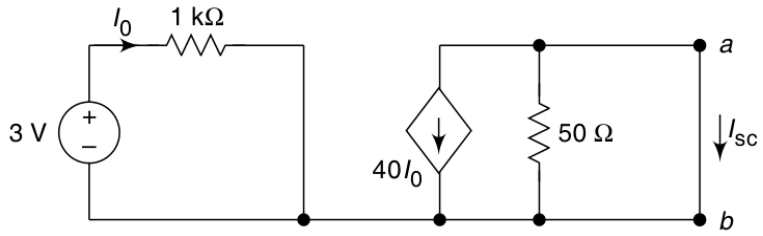
TRY IT FIRST

$V_x = V_{oc}$ feeds back into the left loop. For I_{sc}, $V_x = 0$ removes the $2V_x$ source.

V_{oc} ($R = 50$ kΩ, as in the question figure). Left loop and right node:

$$I_0 = \frac{3 - 2V_x}{1000}, \qquad V_{oc} = V_x = -40I_0 \times 50000 = -2000(3 - 2V_{oc})$$

$$3999V_{oc} = 6000 \qquad \Rightarrow \qquad V_{oc} = 1.5004 \text{ V}$$



a–b shorted

I_{sc} ($V_x = 0 \Rightarrow 2V_x = 0$): $I_0 = \frac{3}{1000} = 3$ mA. The 50 kΩ is shorted, and the dependent source pulls $40I_0$ from a down to b, so the short carries it from b to a:

$$I_{sc}\ (a \to b) = -40I_0 = -0.12 \text{ A}, \qquad R_{th} = \frac{1.5004}{-0.12} = -12.50\ \Omega$$

ANSWER

$V_{th} = 1.500$ V, $R_{th} = -12.50\ \Omega$ (figure value 50 kΩ)

BOOK ERRORS IN 5.27

1. The solution uses **50 Ω** ($V_x = -40I_0 \times 50 = -2000I_0$), not the figure's 50 kΩ. With 50 Ω: $V_{oc} = -2(3 - 2V_{oc}) \Rightarrow V_{oc} = 2$ V.
2. It then writes $R_{th} = 2/0.12 = 16.67\ \Omega$, dropping the sign of $I_{sc} = -0.12$ A. With 50 Ω the correct value is **$R_{th} = -16.67\ \Omega$**.

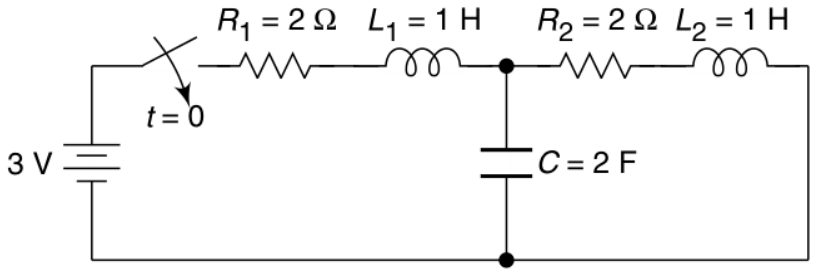
Test-source check (3 V shorted, 1 V applied at a): $I_0 = -\frac{2}{1000}$ A, current drawn $= \frac{1}{R} - 0.08$, giving $R_{th} = -12.50\ \Omega$ (50 kΩ) or $-16.67\ \Omega$ (50 Ω) ✓.

A28 · Problem 5.28

Norton · s-domain transient **new idea**

QUESTION

In the network shown in the figure, the switch is closed at time t = 0. Assuming all the initial currents and voltages as zero, find the current through the inductor L_2 by the use of Norton's theorem.

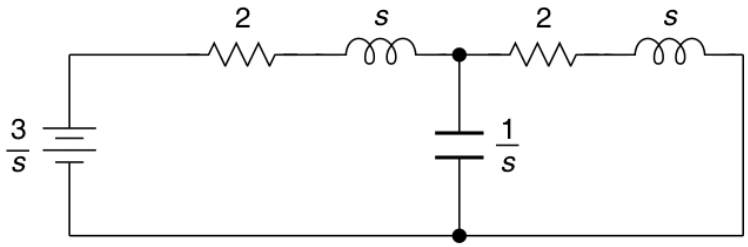


TRY IT FIRST

Draw the s-domain circuit (zero initial conditions). Make (R_2 + sL_2) the load and find the Norton equivalent of the rest.

HINT: WHAT'S NEW

- **Transient = theorem in the s-domain.** A DC source switched on at t = 0 becomes $\frac{V}{s}$; R → R, L → sL, C → $\frac{1}{sC}$. Find $I_L(s)$ exactly as in AC, then invert with partial fractions.
- With A–B shorted the capacitor is shorted out, so I_{sc} is one Ohm's-law step.



The book's s-domain network: note it draws the capacitor as 1/s

s-domain: source $\frac{3}{s}$; $R_1 + sL_1 = s + 2$; C = 2 F → $\frac{1}{2s}$; load $R_2 + sL_2 = s + 2$.

$$Z_N = (s+2) \parallel \frac{1}{2s} = \frac{s+2}{2s^2+4s+1}, \qquad I_N = I_{sc} = \frac{3/s}{s+2} = \frac{3}{s(s+2)}$$

$$I_L(s) = I_N \times \frac{Z_N}{Z_N + (s+2)} = \frac{3}{s(s+2)} \times \frac{1}{1 + (2s^2+4s+1)} = \frac{3}{2s(s+2)(s+1)^2}$$

Partial fractions: $I_L(s) = \frac{3/4}{s} - \frac{3/4}{s+2} + \frac{0}{s+1} - \frac{3/2}{(s+1)^2}$

(Coefficients: at $s = 0$, $\frac{3}{2\cdot2\cdot1} = \frac{3}{4}$; at $s = -2$, $\frac{3}{2(-2)(1)} = -\frac{3}{4}$; on $(s+1)^2$, $\frac{3}{2(-1)(1)} = -\frac{3}{2}$; on $1/(s+1)$, 0, because the s^3 terms must cancel.)

ANSWER

$\mathbf{i_{L2}(t) = \frac{3}{4} - \frac{3}{4}e^{-2t} - \frac{3}{2}te^{-t}\ A}$ $(t \geq 0)$

BOOK ERROR IN 5.28

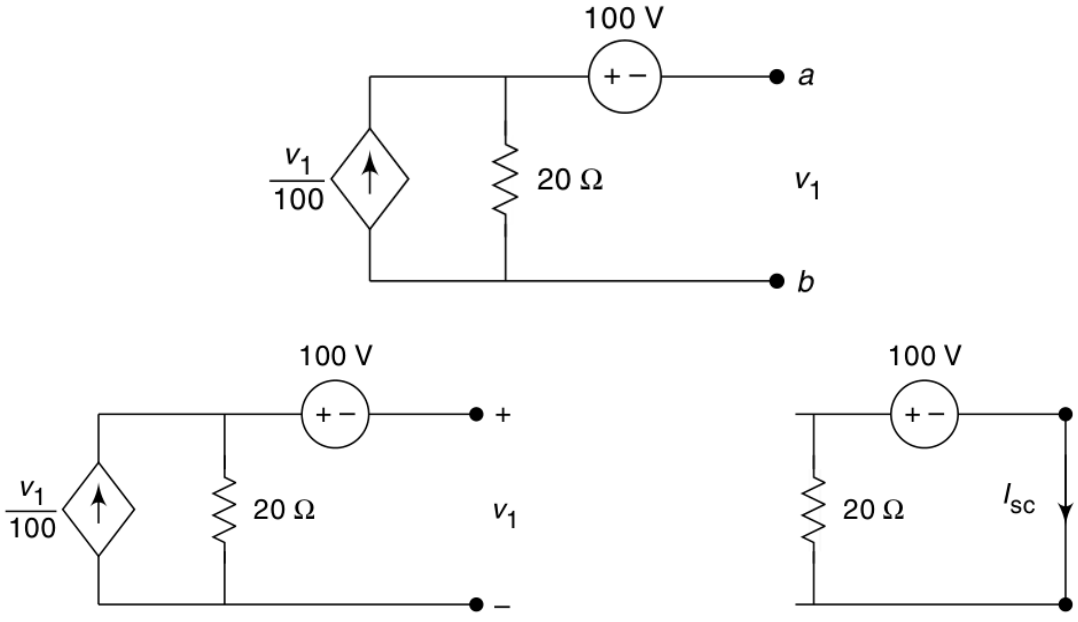
The figure gives C = 2 F, but the book's s-domain network uses $\frac{1}{s}$ (that is C = 1 F) and prints $i(t) = \frac{3}{4} - \frac{3}{4}e^{-2t} - \frac{3}{2}e^{-t}$ sin t A. That answer is right for a 1 F capacitor only. A direct simulation of the circuit equations with C = 2 F matches the answer above (e.g. i(1 s) = 0.0967 A; the book's formula gives 0.184 A). Checks: $i(0) = 0$ and $i(\infty) = \frac{3}{2+2} = 0.75$ A.

A29 · Problem 5.29

Thevenin · dependent source **practice**

QUESTION

The following circuit has a dependent current source and an independent voltage source. Find the Thevenin equivalent network of the circuit across the terminals a and b.



Left: a–b open ($v_1 = v_{oc}$). Right: a–b shorted ($v_1 = 0$, so the dependent source is zero).

V_{oc}: the top node sits at $v_1 + 100$. KCL there:

$$\frac{v_1}{100} = \frac{v_1 + 100}{20} \quad \Rightarrow \quad v_1 = 5v_1 + 500 \quad \Rightarrow \quad V_{oc} = -125\ \text{V}$$

I_{sc}: $v_1 = 0$, so the dependent source is off; the top node is at 100 V and the 20 Ω draws 5 A from it, which must come from the short: I_{sc} (a → b) = −5 A.

$$R_{th} = \frac{-125}{-5} = 25\ \Omega$$

ANSWER

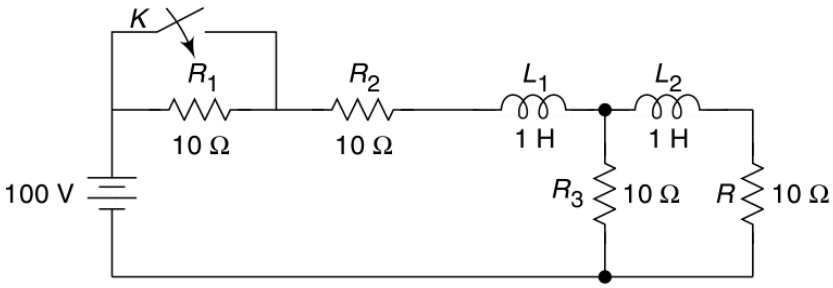
V_{th} = −125 V (i.e. 125 V with + at b), R_{th} = 25 Ω (= book)

A30 · Problem 5.30

Thevenin · transient with initial conditions **new idea**

QUESTION

In the network of the figure, the switch K is closed at time t = 0, a steady state having previously existed. Obtain the current in the resistor R using Thevenin's theorem.



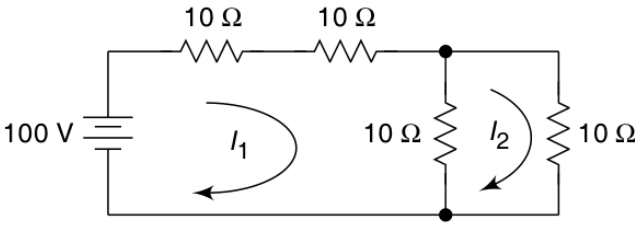
Closing K shorts R_1.

TRY IT FIRST

Find both inductor currents just before t = 0 (inductors are shorts in DC steady state). Then build the s-domain circuit with those currents as sources.

HINT: WHAT'S NEW

- **Non-zero initial current:** an inductor carrying i(0) becomes sL **in series with a voltage source L·i(0)**, polarity aiding the current's direction.



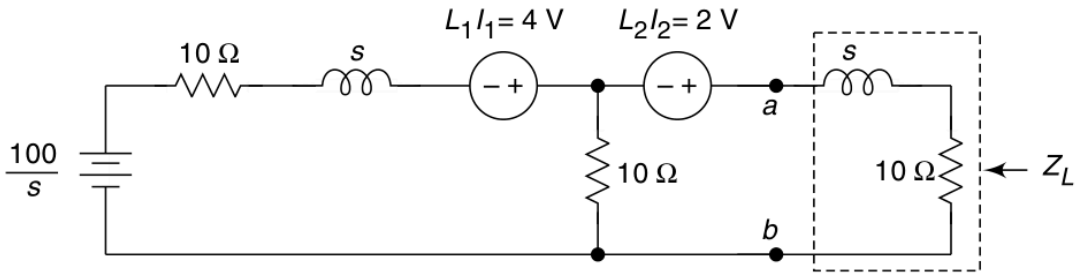
(a) Circuit at $t = 0-$

t = 0−: inductors as shorts

t = 0− (K open, R_1 in):

$$30I_1 - 10I_2 = 100, \qquad -10I_1 + 20I_2 = 0$$

$$i_{L1}(0) = I_1 = 4\ \text{A}, \qquad i_{L2}(0) = I_2 = 2\ \text{A}$$



(b) Transform network for $t > 0$

s-domain network for t > 0 (R_1 shorted); the load $Z_L = s + 10$ is in the dashed box

Thevenin at a–b (load = $sL_2 + R = s + 10$; the 2 V of L_2 stays on the source side):

$$Z_{th} = (s + 10) \parallel 10 = \frac{10(s + 10)}{s + 20}$$

$$I(s) = \frac{100/s + 4}{s + 20} = \frac{4s + 100}{s(s + 20)}, \qquad V_{th} = 10I(s) + 2 = \frac{2s^2 + 80s + 1000}{s(s + 20)}$$

$$I_R(s) = \frac{V_{th}}{Z_{th} + s + 10} = \frac{2s^2 + 80s + 1000}{s(s + 10)(s + 30)} = \frac{10/3}{s} - \frac{2}{s + 10} + \frac{2/3}{s + 30}$$

ANSWER

$$\mathbf{i_R(t) = \frac{10}{3} - 2e^{-10t} + \frac{2}{3}e^{-30t}\ A = 3.33 - 2e^{-10t} + 0.67e^{-30t}\ A}\ (= \text{book})$$

CHECK

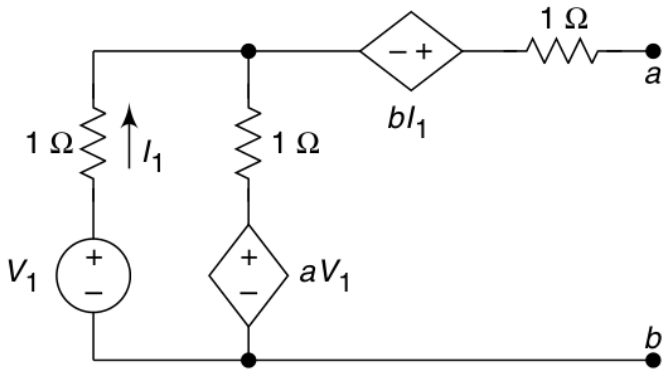
$i(0) = 10/3 - 2 + 2/3 = 2\text{ A} = i_{L2}(0)$ ✓. $i(\infty) = \dfrac{100}{10+5} \times \dfrac{1}{2} = 3.33\text{ A}$ ✓. A direct simulation of the circuit equations agrees at every time tested.

A31 · Problem 5.31

Thevenin · symbolic, two dependent sources **new idea**

QUESTION

For the network shown in the figure, show that the Thevenin equivalent at the terminals a-b is represented by $V_{Th} = \dfrac{V_1}{2}(1 + a + b - ab)$ and $Z_{Th} = \dfrac{3 - b}{2}$.



TRY IT FIRST

Keep a, b and V_1 as letters. Open circuit: no current in the right 1 Ω. Short circuit: two mesh equations.

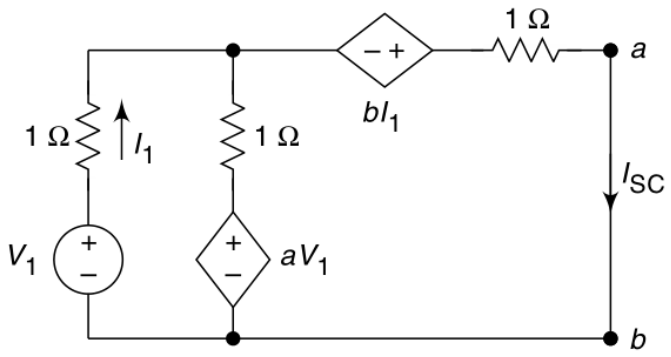
HINT: WHAT'S NEW

- **Symbolic answer:** the method is unchanged; carry the letters and simplify at the end. Substituting numbers (e.g. a = b = 0) checks the algebra.

V_{th} (no current in the right 1 Ω). Left mesh:

$$2I_1 + aV_1 = V_1 \qquad \Rightarrow \qquad I_1 = \frac{V_1}{2}(1 - a)$$

$$V_{th} = aV_1 + 1 \cdot I_1 + bI_1 = aV_1 + (1 + b)\frac{V_1}{2}(1 - a) = \frac{V_1}{2}(1 + a + b - ab)$$



a–b shorted

I_{sc}. Two meshes:

$$2I_1 - I_{SC} = V_1(1 - a) \quad \text{...(i)} \qquad -(1 + b)I_1 + 2I_{SC} = aV_1 \quad \text{...(ii)}$$

$$I_{SC} = \frac{2aV_1 + (1 + b)V_1(1 - a)}{4 - (1 + b)} = \frac{V_1(1 + a + b - ab)}{3 - b}$$

$$Z_{th} = \frac{V_{th}}{I_{SC}} = \frac{3 - b}{2} \qquad \text{(proved)}$$

CHECK

Test source ($V_1 = 0$): the top node sits on $1 \parallel 1 = 0.5\ \Omega$, $I_1 = -0.5I_t$, so $V = I_t + 0.5I_t - 0.5bI_t = \frac{3 - b}{2} I_t$ ✓.

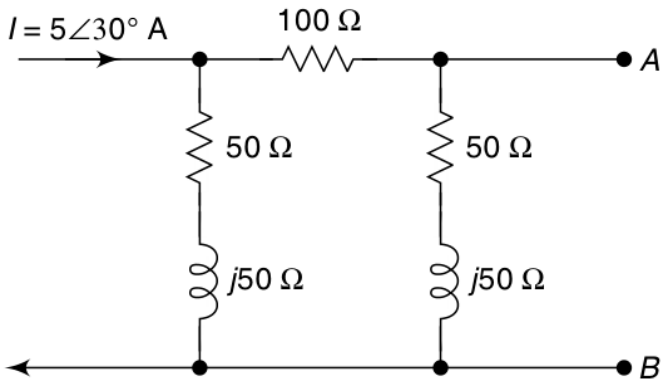
Also confirmed numerically for two sets of (a, b, V_1).

A32 · Problem 5.32

Thevenin · AC, current source **practice**

QUESTION

Find the Thevenin equivalent circuit for the network shown in the figure at terminals A-B.



The 5∠30° A is fed in at the left node and returns on the bottom rail.

$$I_{right} = 5\angle 30° \times \frac{50 + j50}{(100 + 50 + j50) + (50 + j50)} = 5\angle 30° \times \frac{1 + j}{4 + j2} = 1.581\angle 48.43° \text{ A}$$

$$V_{th} = I_{right}(50 + j50) = 111.8\angle 93.43° \text{ V}$$

$$Z_{th} = (150 + j50) \parallel (50 + j50) = \frac{5000 + j10000}{200 + j100} = 40 + j30\ \Omega$$

ANSWER

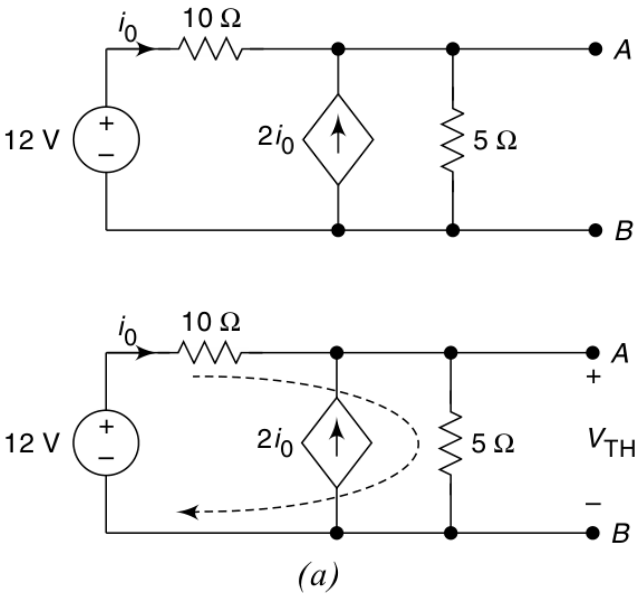
$V_{th} = 111.8\angle 93.43°$ V, $Z_{th} = 50\angle 36.87° = 40 + j30\ \Omega$ (= book)

A33 · Problem 5.33

Thevenin · dependent source **practice**

QUESTION

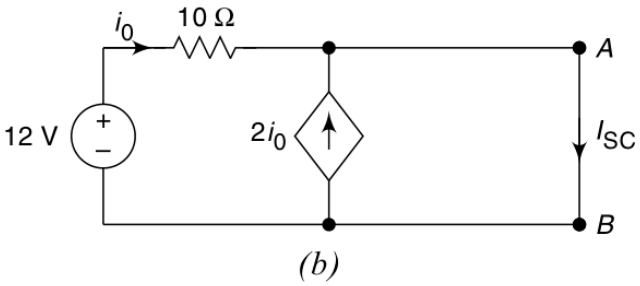
Find Thevenin's equivalent circuit across the terminals A and B for the network shown in the figure.



A–B open

KVL: $10i_0 + V_{th} = 12$ KCL at A: $i_0 + 2i_0 = \frac{V_{th}}{5}$ ⇒ $V_{th} = 15i_0$

$25i_0 = 12$ ⇒ $i_0 = 0.48$ A, $V_{th} = 7.2$ V



A–B shorted (the 5 Ω is shorted out)

$$i_0 = \frac{12}{10} = 1.2\ \text{A}, \qquad I_{sc} = i_0 + 2i_0 = 3.6\ \text{A}, \qquad R_{th} = \frac{7.2}{3.6} = 2\ \Omega$$

ANSWER

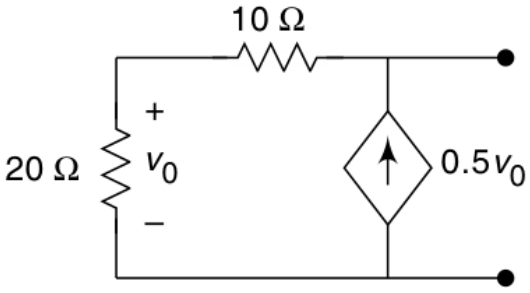
V_{th} = 7.2 V, R_{th} = 2 Ω (= book)

A34 · Problem 5.34

Thevenin · no independent source **new idea**

QUESTION

Find Thevenin's equivalent circuit for the network shown in the figure.

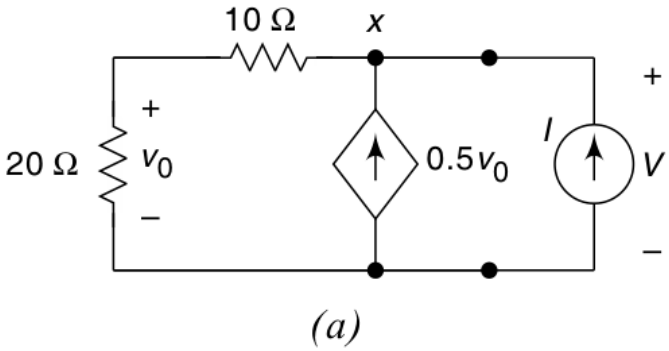


TRY IT FIRST

There is nothing to drive the circuit. What is V_{th}? For Z_{th}, connect a 1 A source and find the voltage it produces.

HINT: WHAT'S NEW

- **Only dependent sources:** V_{th} = 0 (no independent source, no response), and I_{sc} = 0 too, so V_{oc}/I_{sc} fails. Use a **test source**: Z_{th} = V / I.



(a)

Test current I applied; V measured

KCL at x: $\dfrac{V - v_0}{10} - 0.5v_0 - I = 0 \quad \Rightarrow \quad V = 10I + 6v_0$

Current down the 20 Ω = $\dfrac{V - v_0}{10} = 0.5v_0 + I$

$v_0 = 20(0.5v_0 + I) \quad \Rightarrow \quad v_0 = -\dfrac{20}{9}I$

$V = 10I - \dfrac{120}{9}I = -\dfrac{10}{3}I \quad \Rightarrow \quad Z_{th} = -3.33\ \Omega$

ANSWER

$V_{th} = 0$, $Z_{th} = -10/3 = -3.33\ \Omega$ (= book; its final sketch omits the minus sign)